Transmission Line Modelling and Performance

A comprehensive study of line classification, two-port network theory, ABCD parameters, operational phenomena, and structural integrity — the physics that underpins modern power delivery.

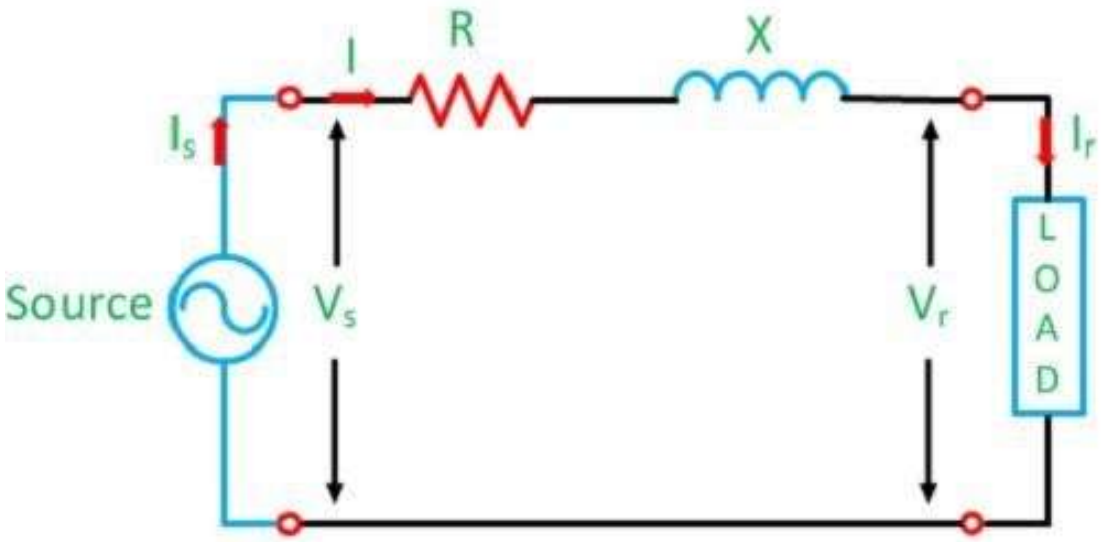
Classification of Transmission Lines

Feature	Short Line	Medium Line	Long Line
Length	≤ 80 km	80–250 km	> 250 km
Typical kV	66–110 kV	132–220 kV	≥ 400 kV
Capacitance	Neglected	Included	Distributed
Model	R + L	Nominal π / T	Long-line model
Charging I	Negligible	Significant	Very significant
Main Use	Local supply	Regional links	Bulk transmission

Representation of Short Transmission Line

Lumped Resistance and Inductance

Capacitance is neglected

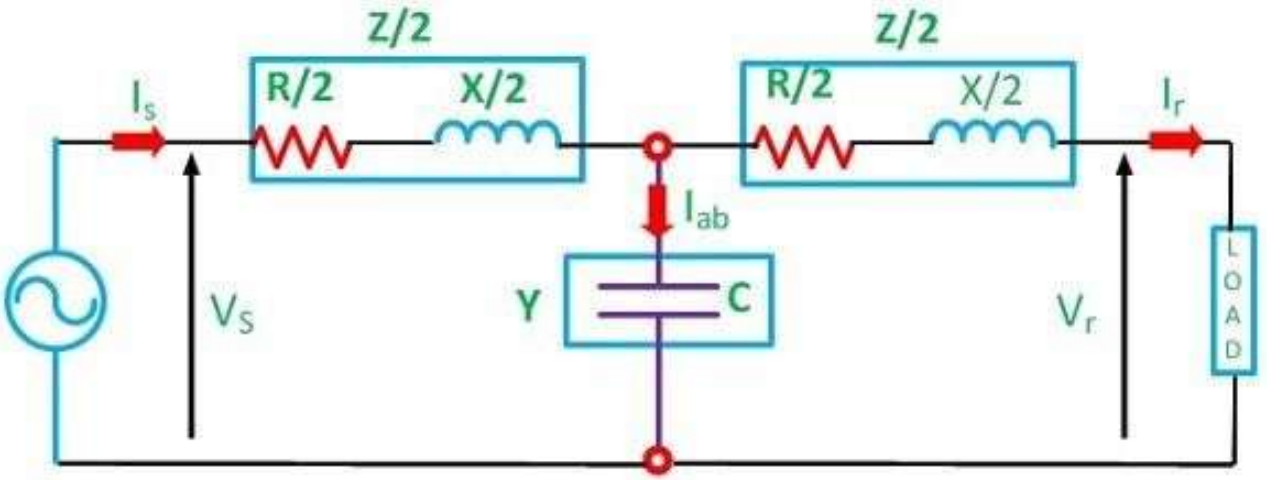


Equivalent circuit model of a short line

Circuit Globe

Representation of a medium transmission line

1. Nominal T

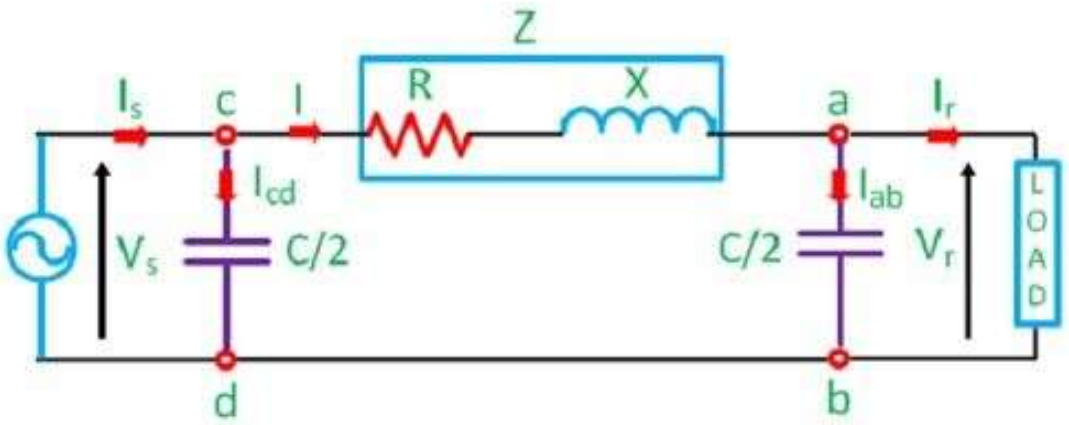


Nominal T model of a Medium Line

Circuit Globe

Representation of a medium transmission line

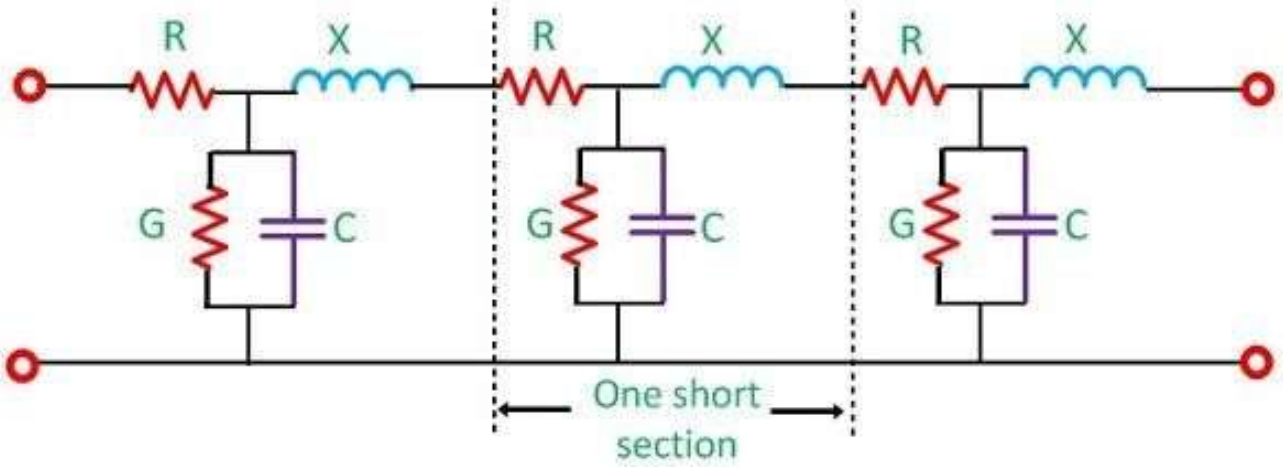
2. Nominal Pi



Nominal Pi model of a medium line.

Circuit Globe

Representation of a long transmission line



Long transmission lines showing distributed parameters.

Circuit Globe

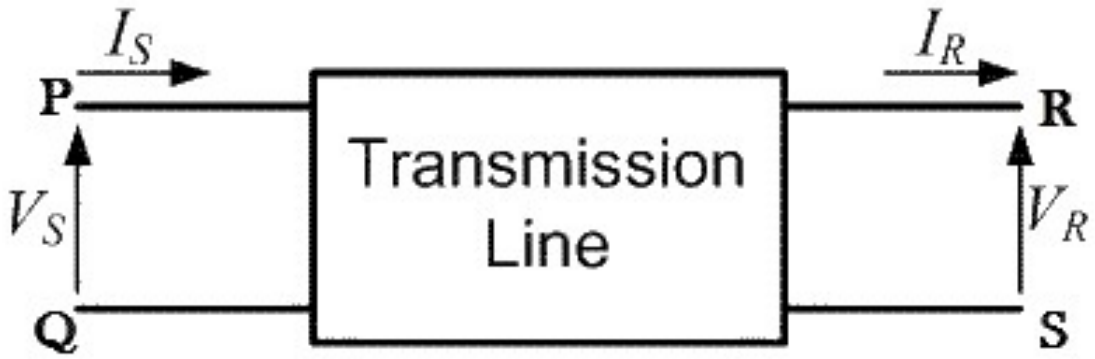
The Transmission Line as a Two-Port Network

Sending End (Port 1)

Voltage **Vs** and current **Is** are the input quantities — representing the generator or grid side of the line.

Receiving End (Port 2)

Voltage **VR** and current **IR** are the output quantities — representing the load side.



The Black-Box Approach

The line's internal complexity is abstracted into four generalised circuit constants — the **ABCD parameters** — which relate the two terminal pairs without needing to specify the internal network topology for every calculation.

This framework applies to any linear, passive, bilateral two-port network.

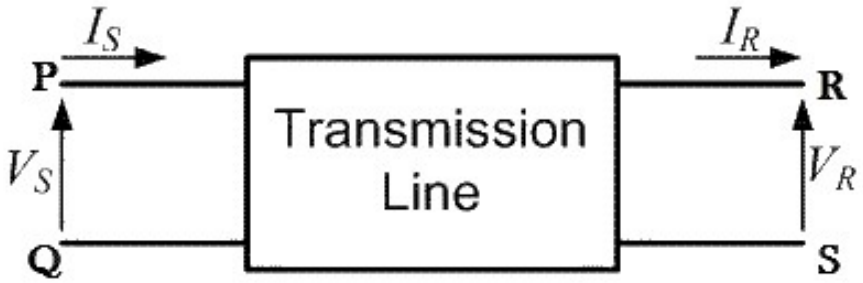
ABCD Parameters: The Analytical Core

The Governing Equations

$$V_S = A \cdot V_R + B \cdot I_R$$

$$I_S = C \cdot V_R + D \cdot I_R$$

Expressed in matrix form, these relate sending-end and receiving-end phasors through a 2×2 transmission matrix.



A — Voltage Ratio

Dimensionless. Ratio of VS to VR at no load.

B — Transfer Impedance

Units: Ω. Links VS to IR.

C — Transfer Admittance

Units: S. Links IS to VR.

D — Current Ratio

Dimensionless. Ratio of IS to IR on short circuit.

Why are ABCD Parameters Needed?

Without ABCD Parameters:

- The line circuit must be analysed whenever the load changes.
- Line parameters must be used repeatedly in every calculation.

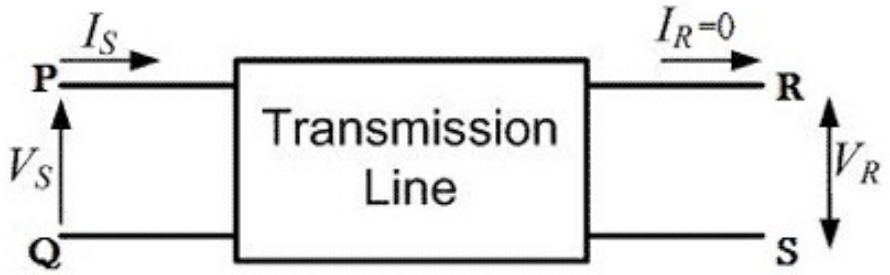
With ABCD Parameters:

Suppose the ABCD constants of a transmission line have already been determined.

- The entire transmission line is represented by four constants: A, B, C and D.

- For any receiving-end voltage and current, the sending-end quantities can be obtained directly without re-analysing the transmission line.

ABCD Parameters When Receiving End is Open Circuited



The receiving end is open-circuited, meaning the receiving end current $I_R = 0$

$$V_S = A\,V_R + B\,0 \Rightarrow V_S = A\,V_R + 0$$

$$A = \left.\frac{V_S}{V_R}\right|_{I_R = 0}$$

$$I_S = C\,V_R + D\,0 \Rightarrow I_S = C\,V_R + 0$$

$$C = \left.\frac{I_S}{V_R}\right|_{I_R = 0}$$

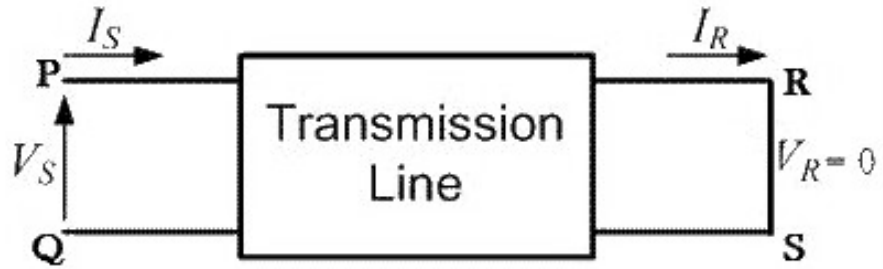
A is the ratio of the sending-end voltage to the open-circuit receiving-end voltage.

A is a dimensionless parameter

C is the open-circuit conductance

The unit of C is mho

ABCD Parameters When Receiving End is Short Circuited



$$V_S = A\,0 + B\,I_R \Rightarrow V_S = 0 + B\,I_R$$

$$B = \left.\frac{V_S}{I_R}\right|_{V_R = 0}$$

$$I_S = C\,0 + D\,I_R \Rightarrow I_S = 0 + D\,I_R$$

$$D = \left.\frac{I_S}{I_R}\right|_{V_R = 0}$$

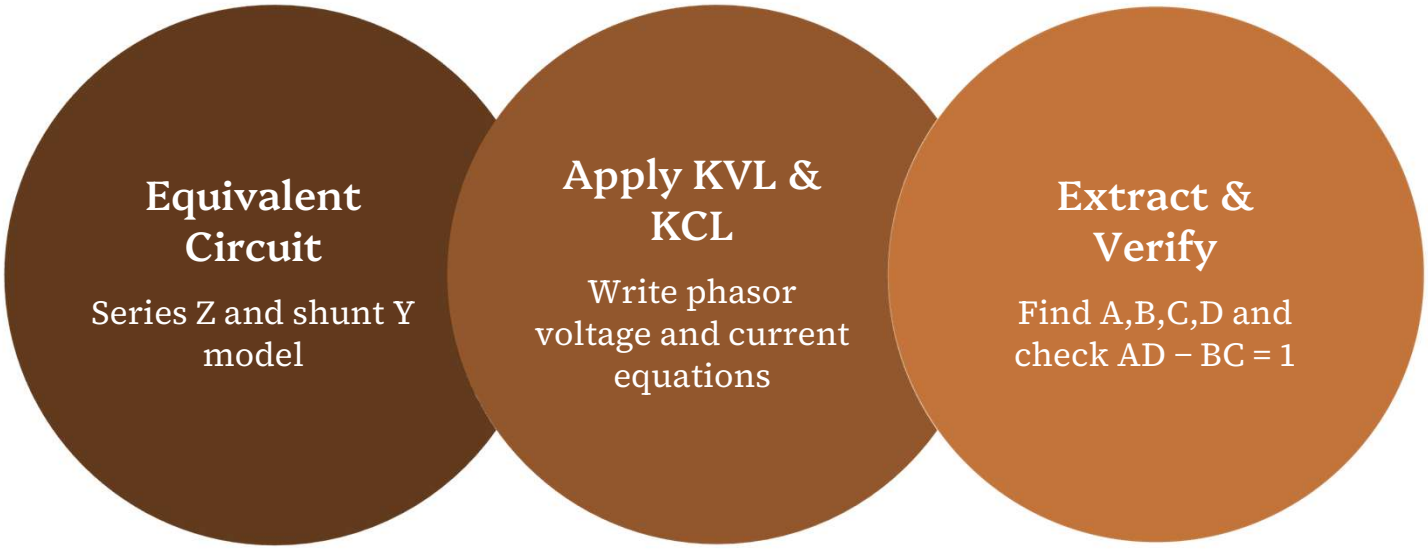
B is the ratio of the sending-end voltage to the short-circuit receiving-end current.

Since dimension-wise B is a ratio of voltage to current, its unit is Ω.

D is the ratio of sending-end current to the short-circuit receiving-end current.

D is a ratio of current to current; it's a dimensionless parameter

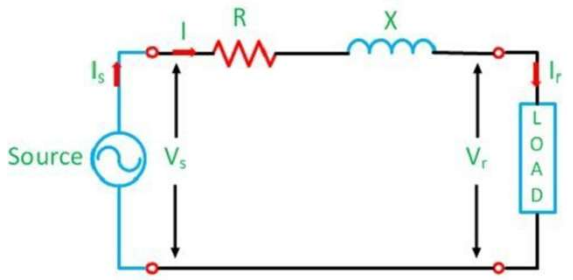
Deriving the ABCD Parameters



The derivation begins by choosing an appropriate equivalent circuit for the line class. Kirchhoff's laws yield simultaneous phasor equations, and the reciprocity constraint **AD − BC = 1** serves as a vital verification check — any deviation signals an error in the circuit model.

Short Lines (Simple Series Impedance Method)

AD – BC = 1



Equivalent circuit model of a short line

Circuit Globe

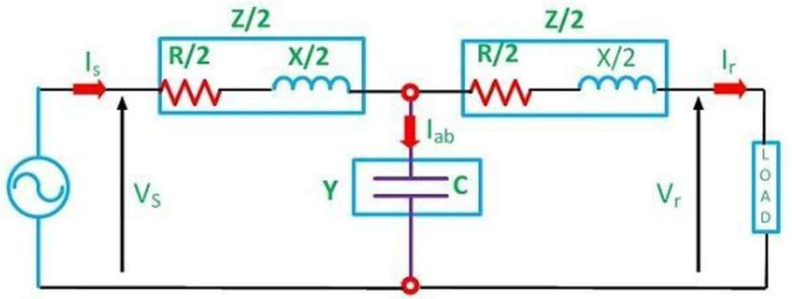
$$\mathbf{I_S} = \mathbf{I_R} \qquad \ldots(6.43)$$

$$\text{and} \quad \mathbf{V_S} = \mathbf{V_R} + \mathbf{I_R Z} \qquad \ldots(6.44)$$

Comparing Eq. (6.44) with Eq. (6.35) and Eq. (6.43) with Eq. (6.36), we have

$$\mathbf{A} = 1;\ \mathbf{B} = \mathbf{Z};\ \mathbf{C} = 0 \quad \text{and} \quad \mathbf{D} = 1$$

Nominal-T Method



Nominal T model of a Medium Line

Circuit Globe

AD – BC = 1

$$\mathbf{V}_S = \mathbf{V}' + \mathbf{I}_S \frac{\mathbf{Z}}{2}$$

$$\text{and } \mathbf{V}' = \mathbf{V}_R + \mathbf{I}_R \frac{\mathbf{Z}}{2}$$

$$\text{and } \mathbf{I}_S = \mathbf{I}_R + \mathbf{I}_C$$

$$= \mathbf{I}_R + \mathbf{Y}\mathbf{V}'$$

$$= \mathbf{I}_R + \mathbf{Y}\left(\mathbf{V}_R + \mathbf{I}_R \frac{\mathbf{Z}}{2}\right)$$

$$= \mathbf{Y}\mathbf{V}_R + \mathbf{I}_R\left(1 + \frac{\mathbf{YZ}}{2}\right) \qquad \ldots(6.49)$$

Substituting the value of $\mathbf{V}'$ from Eq. (6.48) in Eq. (6.47), we have

$$\mathbf{V}_S = \mathbf{V}_R + \mathbf{I}_R \frac{\mathbf{Z}}{2} + \mathbf{I}_S \frac{\mathbf{Z}}{2}$$

$$= \mathbf{V}_R + \mathbf{I}_R \frac{\mathbf{Z}}{2} + \frac{\mathbf{Z}}{2}\left[\mathbf{Y}\mathbf{V}_R + \mathbf{I}_R\left(1 + \frac{\mathbf{YZ}}{2}\right)\right]$$

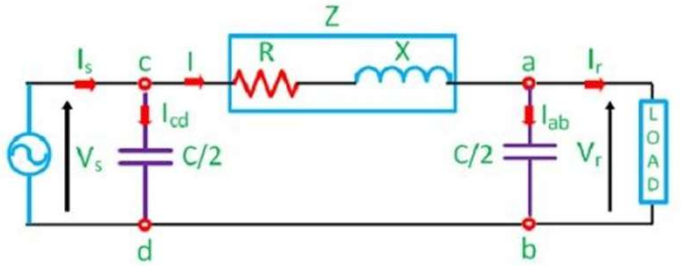
$$= \left(1 + \frac{\mathbf{YZ}}{2}\right)\mathbf{V}_R + \left(\mathbf{Z} + \frac{\mathbf{YZ}^2}{4}\right)\mathbf{I}_R \qquad \ldots(6.50)$$

Comparing Eq. (6.50) with Eq. (6.35) and Eq. (6.49) with Eq. (6.36), we have

$$\mathbf{A} = \mathbf{D} = 1 + \frac{\mathbf{YZ}}{2}; \mathbf{B} = \mathbf{Z}\left(1 + \frac{\mathbf{YZ}}{4}\right) \text{ and } \mathbf{C} = \mathbf{Y}$$

Nominal-π Method

AD – BC = 1



Nominal Pi model of a medium line.

Circuit Globe

$$\mathbf{I}_L = \mathbf{I}_R + \frac{\mathbf{Y}}{2}\mathbf{V}_R \qquad \ldots(6.51)$$

and $$\mathbf{I}_S = \mathbf{I}_L + \frac{\mathbf{Y}}{2}\mathbf{V}_S \qquad \ldots(6.52)$$

and $$\mathbf{V}_S = \mathbf{V}_R + \mathbf{I}_L\mathbf{Z} \qquad \ldots(6.53)$$

Substituting $\mathbf{I}_L = \mathbf{I}_R + \frac{1}{2}\mathbf{Y}\mathbf{V}_R$ from Eq. (6.51), we have

$$\mathbf{V}_S = \mathbf{V}_R + \mathbf{Z}\left(\mathbf{I}_R + \frac{1}{2}\mathbf{Y}\mathbf{V}_R\right)$$

$$= \mathbf{V}_R\left(1 + \frac{1}{2}\mathbf{Y}\mathbf{Z}\right) + \mathbf{I}_R\mathbf{Z} \qquad \ldots(6.54)$$

From Eqs. (6.51) and (6.52), we have

$$\mathbf{I}_S = \mathbf{I}_R + \frac{1}{2}\mathbf{Y}\mathbf{V}_R + \frac{1}{2}\mathbf{Y}\mathbf{V}_S \qquad \ldots(6.55)$$

$$\mathbf{I}_S = \mathbf{I}_R + \frac{1}{2}\mathbf{Y}\mathbf{V}_R + \frac{1}{2}\mathbf{Y}\left[\left(1 + \frac{1}{2}\mathbf{Y}\mathbf{Z}\right)\mathbf{V}_R + \mathbf{I}_R\mathbf{Z}\right]$$

$$= \mathbf{V}_R\mathbf{Y}\left(1 + \frac{1}{4}\mathbf{Y}\mathbf{Z}\right) + \mathbf{I}_R\left(1 + \frac{1}{2}\mathbf{Y}\mathbf{Z}\right) \qquad \ldots(6.56)$$

Comparing Eq. (6.54) with Eq. (6.35) and Eq. (6.56) with Eq. (6.36), we have

$$\mathbf{A} = \mathbf{D} = 1 + \frac{1}{2}\mathbf{Y}\mathbf{Z};\ \mathbf{B} = \mathbf{Z} \text{ and } \mathbf{C} = \mathbf{Y}\left(1 + \frac{1}{4}\mathbf{Y}\mathbf{Z}\right)$$

To find the parameters given Z and Y

A and D → Same for both models : $A = D = 1 + YZ/2$

B & C changes

T : $B = Z(1 + YZ/4)$, $C = Y$

π : $B=Z$, $C = Y(1 + YZ/4)$

So, $BC = YZ(1 + YZ/4)$ for both. (Rem as TC=Y?)

A, D – half, BC- quarter

AD-BC = 1 for both

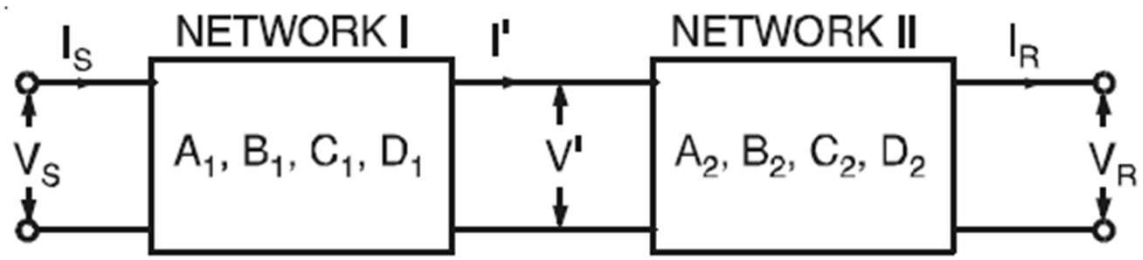


Fig. 6.27. *Two Networks in Tandem*

$$\mathbf{A} = \mathbf{A}_1\mathbf{A}_2 + \mathbf{B}_1\mathbf{C}_2$$
$$\mathbf{B} = \mathbf{A}_1\mathbf{B}_2 + \mathbf{B}_1\mathbf{D}_2$$
$$\mathbf{C} = \mathbf{A}_2\mathbf{C}_1 + \mathbf{C}_2\mathbf{D}_1$$
$$\mathbf{D} = \mathbf{B}_2\mathbf{C}_1 + \mathbf{D}_1\mathbf{D}_2$$

$$T_1 = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \qquad T_2 = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \qquad T = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$T = T_1 T_2$$

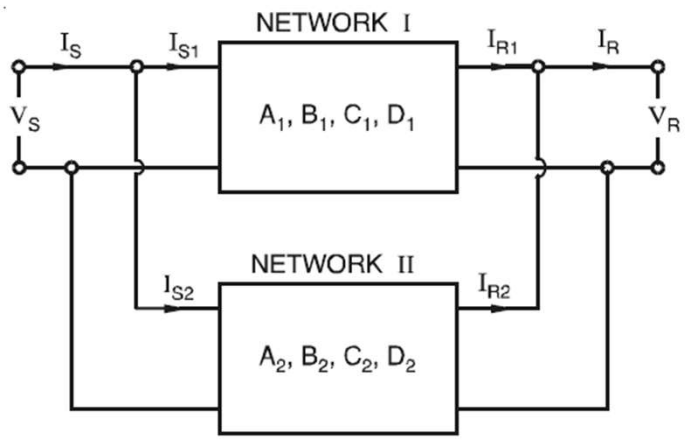


Fig. 6.28. *Two Networks in Parallel*

$$\mathbf{A} = \frac{\mathbf{A_1 B_2 + A_2 B_1}}{\mathbf{B_1 + B_2}}$$

$$\mathbf{B} = \frac{\mathbf{B_1 B_2}}{\mathbf{B_1 + B_2}}$$

$$\mathbf{C} = \mathbf{C_1 + C_2} - \frac{(\mathbf{A_1 - A_2})(\mathbf{D_1 - D_2})}{\mathbf{B_1 + B_2}}$$

$$\mathbf{D} = \frac{\mathbf{B_1 D_2 + B_2 D_1}}{\mathbf{B_1 + B_2}}$$

$$\text{Denominator} = B_1 + B_2$$

$$A = \frac{A_1 B_2 + A_2 B_1}{B_1 + B_2}$$

B has only B in its equation

Calculate C using AD-BC = 1

$$D = \frac{D_1 B_2 + D_2 B_1}{B_1 + B_2}$$

Replace A by D, and the equation remains identical

Example 6.23. **Find the A, B, C, D parameters of a 3-phase 80 km 50 Hz transmission line with series impedance of 0.15 + j0.78 ohm per km and a shunt capacitance of $j5 \times 10^{-6}$ mho per km.** [*Pb. Technical Univ. Power System-I (T&D) May 2008*]

Solution : Total series impedance of the line,

$$\mathbf{Z} = (0.15 + j0.78) \times 80 = (12 + j62.4)\ \Omega$$

Total shunt capacitance of the line

$$\mathbf{Y} = j5 \times 10^{-6} \times 80 = j4 \times 10^{-4} \text{ mho}$$

In nominal-T configuration

$$\mathbf{A} = \mathbf{D} = 1 + \frac{1}{2}\mathbf{YZ}$$

$$= 1 + \frac{1}{2}(12 + j62.4) \times j4 \times 10^{-4}$$

$$= (0.98752 + j0.0024) \quad \textbf{Ans.}$$

$$\mathbf{B} = \mathbf{Z} \times \left(1 + \frac{\mathbf{YZ}}{4}\right)$$

$$= (12 + j62.4)\left[1 + \frac{1}{4} \times j4 \times 10^{-4} \times (12 + j62.4)\right]$$

$$= (12 + j62.4)(0.99376 + j0.0012)$$

$$= (11.85024 + j11.91072) \text{ ohms } \textbf{Ans.}$$

$$\mathbf{C} = \mathbf{Y} = j4 \times 10^{-4} = 0.0004 \text{ S } \textbf{Ans.}$$

Example 6.24. The A, B, C, D constants of a three-phase transmission line are $A = D = 0.936 + j0.016$, $B = 33.5 + j138\ \Omega$ and $C = (-0.9280 + j901.223) \times 10^{-6}$ mho.

The load at the receiving end is 40 MW at 200 kV with pf of 0.86 lagging. Find the magnitude of the sending-end voltage, current, power, and voltage regulation. Assume that the magnitude of the sending-end voltage remains constant.

Solution : Receiving-end phase voltage,

$$V_R = \frac{200 \times 1{,}000}{\sqrt{3}} = 115{,}470 \text{ V}$$

Receiving-end current,

$$I_R = \frac{40 \times 10^6}{\sqrt{3} \times 200 \times 10^3 \times 0.86} = 134.27 \text{ A}$$

Power factor, $\cos \phi = 0.86$

Phase angle, $\phi = \text{Cos}^{-1}\, 0.86 = -30.68°$

$\sin \phi = 0.51$

$$\mathbf{A} = \mathbf{D} = 0.936 + j0.016 = 0.93614 \angle 0.98°$$

$$\mathbf{B} = 33.5 + j138 = 142 \angle 76.355° \ \Omega$$

$$\mathbf{C} = (-0.9280 + j901.223) \times 10^{-6}$$

$$= 901.2235 \angle 90.06° \times 10^{-6} \ \text{℧}$$

Taking receiving-end phase voltage as reference phasor, we have

$$\mathbf{V}_R = 115{,}470(1 + j0) = 115{,}470 \angle 0^\circ \text{ V}$$

$$\mathbf{I}_R = 134.27 \angle -30.68^\circ \text{ A}$$

Sending-end phase voltage,

$$\mathbf{V}_S = \mathbf{A}\mathbf{V}_R + \mathbf{B}\mathbf{I}_R$$

$$= 0.93614 \angle 0.98^\circ \times 115{,}470 \angle 0^\circ$$

$$+ 142 \angle 76.355^\circ \times 134.27 \angle -30.68^\circ$$

$$= 108{,}096 \angle 0.98^\circ + 19{,}066 \angle 45.675^\circ$$

$$= (108{,}080 + j1848.8) + (13{,}322 + j13{,}639.6)$$

$$= (121{,}402 + j15{,}488.4) = 122{,}386 \angle 7.27^\circ \text{ V}$$

Sending-end line voltage,

$$V_{SL} = \sqrt{3} \times V_S = \sqrt{3} \times 122{,}386$$

$$= 211{,}979 \text{ V or } 211.98 \text{ kV} \quad \textbf{Ans.}$$

Sending-end current,

$$\mathbf{I}_S = \mathbf{C}\mathbf{V}_R + \mathbf{D}\mathbf{I}_R$$

$$= 104.064 \angle 90.06^{\circ} + 125.696 \angle -29.7^{\circ}$$

$$= (-0.109 + j104.064) + (109.18 - j62.28)$$

$$= (109.071 + j41.784) = 116.8 \angle 20.96^{\circ} \text{ A} \quad \textbf{Ans.}$$

Sending-end phase angle,

$$\phi_S = 20.96^{\circ} - 7.27^{\circ} = 13.69^{\circ} = 0.9716 \text{ (leading)} \quad \textbf{Ans.}$$

Sending-end power

$$= \sqrt{3} V_{SL} I_S \times \cos \phi_S \times 10^{-6} \text{ MW}$$

$$= \sqrt{3} \times 211.98 \times 10^3 \times 116.8 \times 0.9716 \times 10^{-6}$$

$$= 41.666 \text{ MW} \quad \textbf{Ans.}$$

Voltage regulation

$$= \frac{V_{SL} - V_{RL}}{V_{RL}} \times 100 = \frac{211.98 - 200}{200} \times 100 = 5.99\% \quad \textbf{Ans.}$$

Calculating Parameters by Line Type

Short Line

A = D = 1
B = Z
C = 0

Shunt admittance ignored. Only series impedance Z = R + jωL retained.

Medium Line — Nominal T

A = D = 1 + ZY/2
B = Z(1 + ZY/4)
C = Y

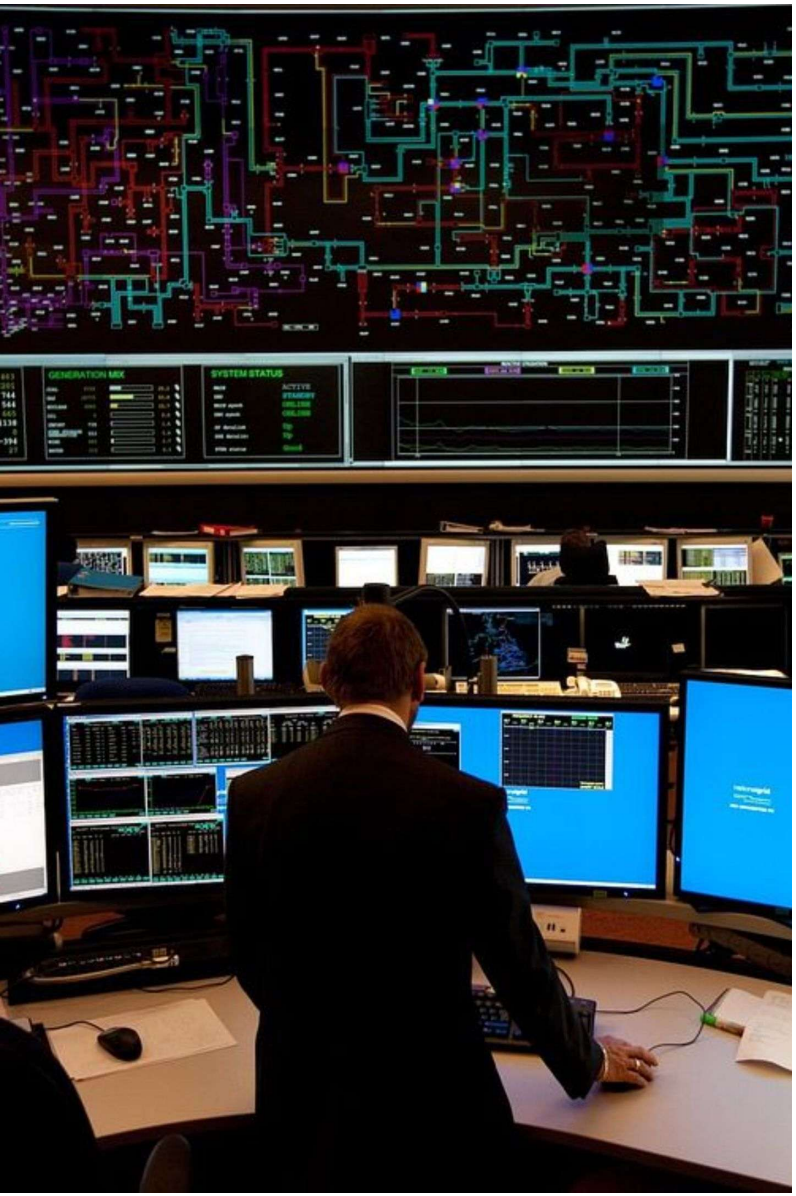
Impedance split equally; total shunt admittance placed at the centre.

Medium Line — Nominal π

A = D = 1 + ZY/2
B = Z
C = Y(1 + ZY/4)

Total series impedance at centre; admittance split equally at both ends.

All calculations involve complex number arithmetic — accurate phasor representation is essential for correct voltage regulation and efficiency results.



CHAPTER 2

Operational Phenomena

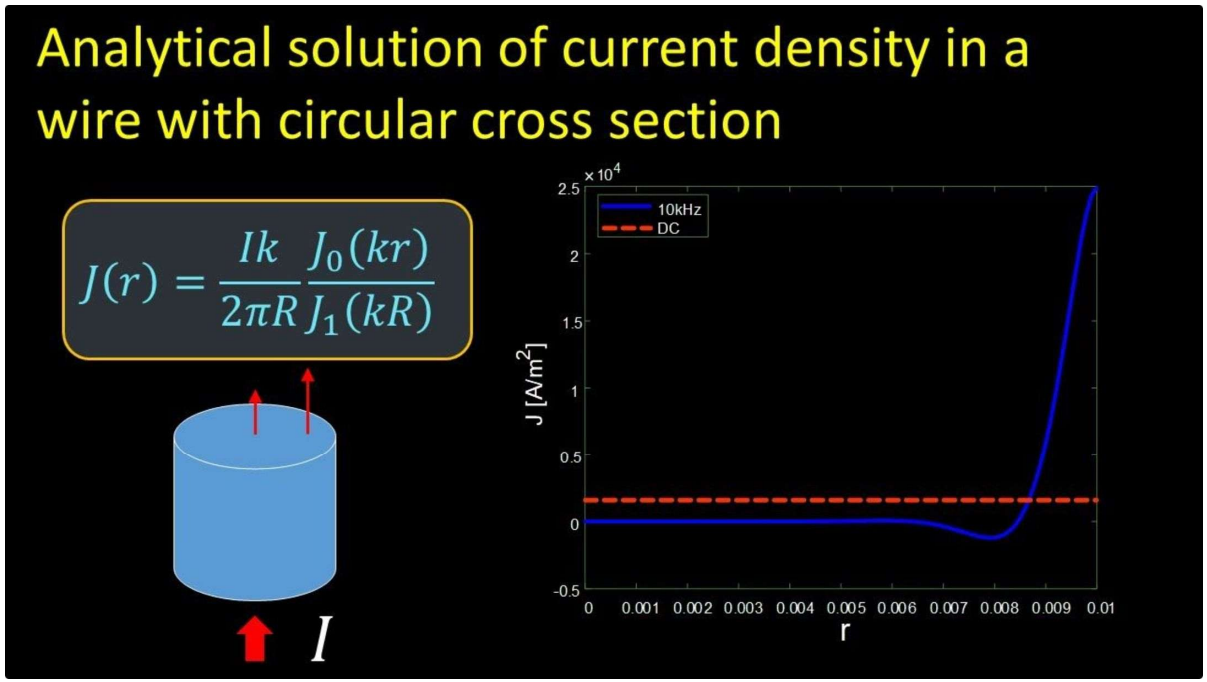
Voltage Regulation

Percentage rise in receiving-end voltage from full-load to no-load — a key stability metric.

Transmission Efficiency

Ratio of power delivered at the receiving end to the power sent — directly impacts operating costs.

The Skin Effect



At AC frequencies, current does not distribute uniformly across a conductor's cross-section. Instead, it concentrates in a thin layer near the **surface** — the "skin" — due to opposing eddy currents in the core.

- **Skin depth** decreases as frequency increases — at 50 Hz it is typically a few millimetres in copper.
- **Effective resistance** rises because less cross-sectional area carries current, increasing I^2R losses.
- Bundled and stranded conductors (e.g. ACSR) are designed to partially mitigate this effect.

SKIN EFFECT

- For DC, current is uniformly distributed over the entire cross-section of the conductor.
- For AC, current is not uniformly distributed and tends to concentrate near the outer surface of the conductor.
- Consequently, the effective conducting area decreases, increasing the effective AC resistance.

Skin effect is the tendency of alternating current to concentrate near the surface of a conductor, increasing its effective resistance.

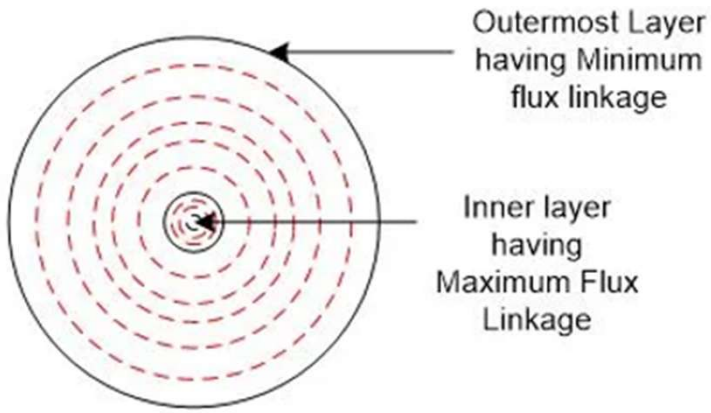


Fig. : Cross-sectional view of a conductor made up of several concentric Cylinders

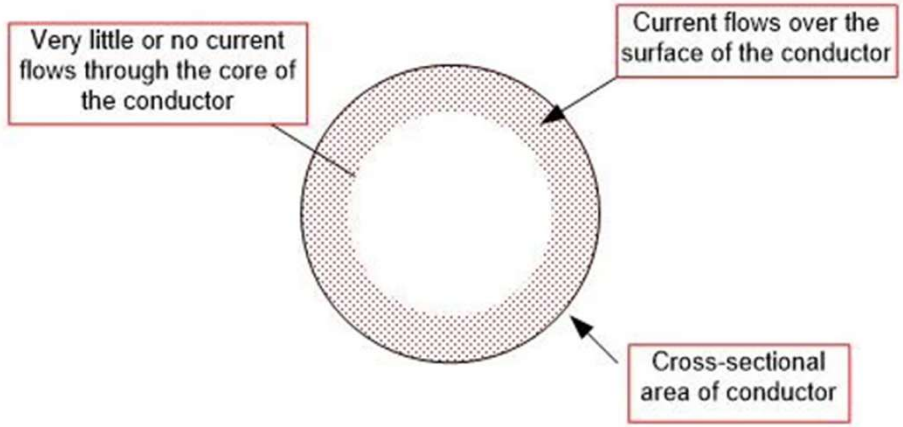


Fig. : Cross-sectional view of a conductor having Skin Effect

Principle Behind Skin Effect

Consider a solid conductor as consisting of a large number of current-carrying filaments.

- Each filament produces magnetic flux around it.
- Inner filaments are linked by more magnetic flux than outer filaments.
- Hence, inner filaments have higher inductive reactance.
- Current naturally flows through paths of lower reactance.

Result

Current density becomes maximum near the surface and minimum at the centre of the conductor, producing a non-uniform current distribution.

Factors Affecting Skin Effect

Skin effect increases with:

- Higher operating frequency
- Larger conductor diameter
- Higher magnetic permeability of the conductor material
- Solid conductor construction

Observations

- Skin effect is negligible for DC and very small at low frequencies.
- It becomes significant in large solid conductors and at high frequencies.

Practical Significance of Skin Effect

Effects

- Increases the effective AC resistance of the conductor.
- Increases I^2R losses and slightly reduces transmission efficiency.

Methods to Reduce Skin Effect

- Use stranded conductors instead of solid conductors.
- Use hollow conductors for large bus-bars.
- Use conductors with low magnetic permeability.

In transmission lines, stranded ACSR conductors are preferred because they experience much less skin effect than solid conductors.

Ferranti Effect

The Paradox of No-Load Long Lines

Under light or no-load conditions, the receiving-end voltage of a long line can **exceed** the sending-end voltage — an apparently counterintuitive result.

Mechanism: The line's distributed capacitance draws a leading **charging current** even with no load connected. This current flowing through the line's inductive reactance produces a voltage *rise* towards the receiving end.

Why It Matters

Equipment Overvoltage

Transformers and switchgear must be rated for the elevated no-load voltage.

Mitigation

Shunt reactors are connected at the receiving end to absorb excess reactive power and suppress voltage rise.

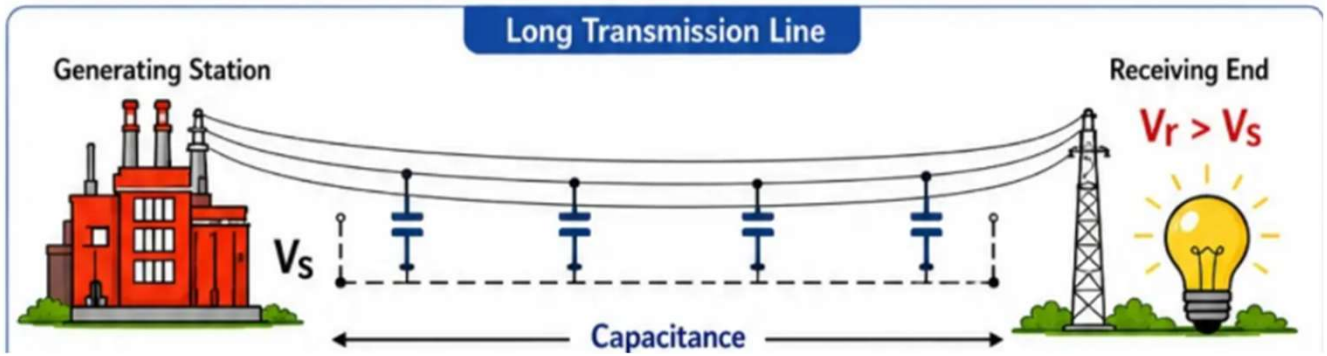
FERRANTI EFFECT

Definition

The Ferranti Effect is the phenomenon in which the receiving-end voltage of a medium- or long-transmission line exceeds the sending-end voltage when the line is open-circuited or lightly loaded.

Characteristics

- Occurs mainly in medium and long transmission lines.
- Negligible in short transmission lines due to their small capacitance.
- Caused by the combined effect of line capacitance and line inductance.
- The receiving-end voltage may be several percent higher than the sending-end voltage under no-load conditions.

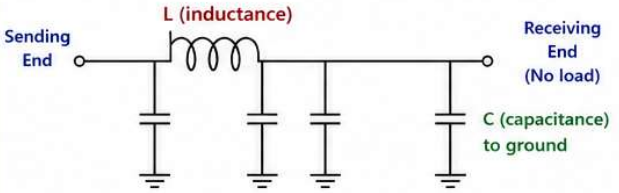


FERRANTI EFFECT – INTUITIVE EXPLANATION

On a long transmission line under light load or no load, the receiving-end voltage becomes **higher than** the sending-end voltage.

1. WHY DOES IT HAPPEN? – PHYSICAL PICTURE

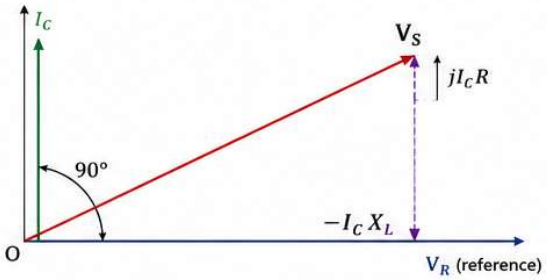
A transmission line has **inductance in series** (L) and **capacitance in shunt** (C) distributed along its length.



1. The shunt capacitance draws a **leading (capacitive) charging current** I_C even when the line supplies no real power.
2. This current flows through the **series inductance** L of the line.
3. Whenever a current changes, the inductor produces a **back emf** that opposes the change in that current (Lenz's law).
4. Because I_C is leading, the back emf produced by L has a direction such that, when we move from sending end to receiving end, it **acts in the same direction as** V_R. Hence, the **voltage rises**.

The inductor is not "helping" or "reversing" its nature.
It still **opposes the change in current**.
Because the charging current is leading, the induced voltage due to L results in a **rise of voltage** toward the receiving end.

2. PHASOR VIEW (No load / Light load)



- I_C leads V_R by 90°.
- Voltage across inductance: $j I_C X_L$ is 180° from V_R (acts in **opposite direction to** V_R from the sending end).
- Adding this to V_R, the resultant sending-end voltage V_S is smaller in magnitude.
- Hence, $V_R > V_S$ ⇒ **Ferranti Effect (voltage rise)**

3. THE BACK EMF – KEY INTUITION

Inductor law: $e_L = -L\dfrac{di}{dt}$

The inductor always produces a voltage that **opposes the change in current**.

Here, the change in current is the capacitive charging current I_C.

Due to the phase of I_C, the back emf due to L results in a **voltage rise** toward the receiving end.



4. FACTORS AFFECTING FERRANTI RISE

- Increases with **line length** (more L and more C).
- Increases with **system voltage**.
- More pronounced at **light load or no load**.
- Decreases with **increase in load**.

5. TAKEAWAY

Ferranti effect is not due to any "mysterious reversal" of inductance. It is due to the **leading capacitive current** from line capacitance, which, through the line inductance, produces a back emf that **raises the receiving-end voltage**.

6. APPROXIMATE EXPRESSION (No load, R neglected)

For a long line with total inductance L and total shunt capacitance C,

$$V_S = V_R - \underbrace{\omega^2 LC\, V_R}_{\text{Voltage drop due to inductance acting on charging current}} \Rightarrow \boxed{V_R \approx V_S\,(1 + \omega^2 LC)}$$

Ferranti rise

Approximate % rise $= \omega^2 LC \times 100$

Ferranti Effect: At light load or no load, $V_R > V_S$ due to the back emf of line inductance produced by the leading charging current.

Why Does the Ferranti Effect Occur?

Under no-load or light-load conditions:

- The line current is dominated by the capacitive charging current (I_C)

- Since I_C leads the receiving-end voltage (V_R) by nearly 90°, the voltage across the line inductive reactance (jI_CX_L) is reversed.

- Instead of opposing the applied voltage as in normal operation, the inductive voltage assists it.

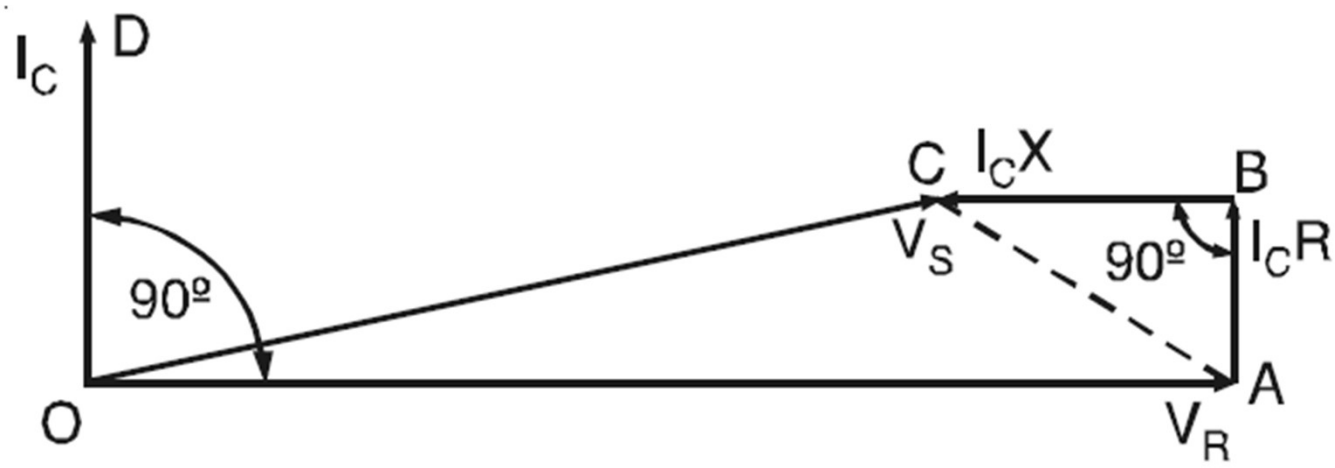
Result

- The line experiences a voltage rise instead of a voltage drop.

- Consequently, the receiving-end voltage becomes greater than the sending-end voltage.

$$V_R > V_S$$

Phasor diagram



Mathematical Explanation

$V_S = V_R + jI_CR - I_CX_L$

Neglecting the line resistance,

$V_S = V_R - I_CX_L$

Since

$I_C = j\omega CV_R$

Substituting,

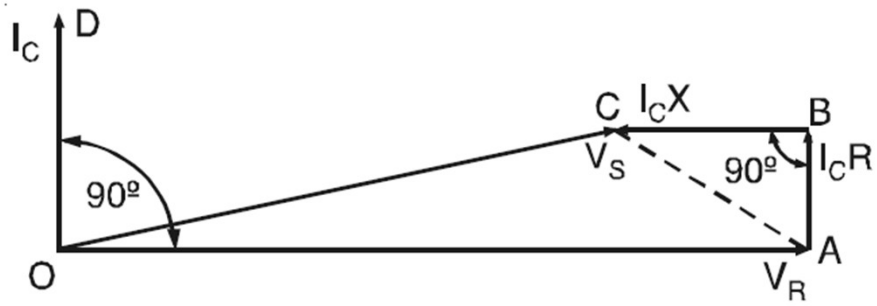
$V_S = V_R - \omega^2 LCV_R$

Hence,

$V_R > V_S$

Approximate rise in receiving-end voltage

Rise in Voltage = $(\omega^2 l^2 V_R \times 10^{-10})/18$



Observations

- Voltage rise is proportional to the square of the line length.
- Therefore, the Ferranti Effect becomes significant only in long transmission lines.

$$C = C_0 l$$

$$\vec{V}_S = \vec{V}_R + jI_C R - I_C X_L$$

$$\Delta V = V_R - V_S = I_C X_L$$

Therefore, the average charging current is

$$I_C = \frac{1}{2}\omega C V_R$$

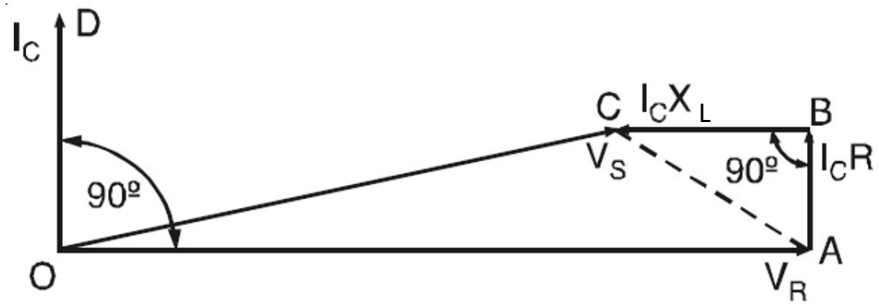
Neglecting resistance,

$$V_S \approx V_R - I_C X_L$$

$$\Delta V = \left(\frac{1}{2}\omega C_0 l V_R\right)(\omega L_0 l)$$

$$I_C = \frac{1}{2}\omega C_0 l V_R$$

$$\boxed{\Delta V = \frac{1}{2}\omega^2 L_0 C_0 l^2 V_R}$$



$$X_L = \omega L_0 l$$

$$\boxed{V_R - V_S = \frac{1}{2}\omega^2 L_0 C_0 l^2 V_R}$$

$$\frac{V_R - V_S}{V_R} = \frac{1}{2}\frac{X_L}{X_C}$$

$$V_R - V_S = \frac{1}{2}\omega^2 L_0 C_0 l^2 V_R$$

$$\boxed{\frac{V_R - V_S}{V_R} = \frac{1}{2}\omega^2 L_0 C_0 l^2}$$

For an overhead line,

$$\frac{1}{\sqrt{L_0 C_0}} = 3 \times 10^8 \text{ m/s}$$

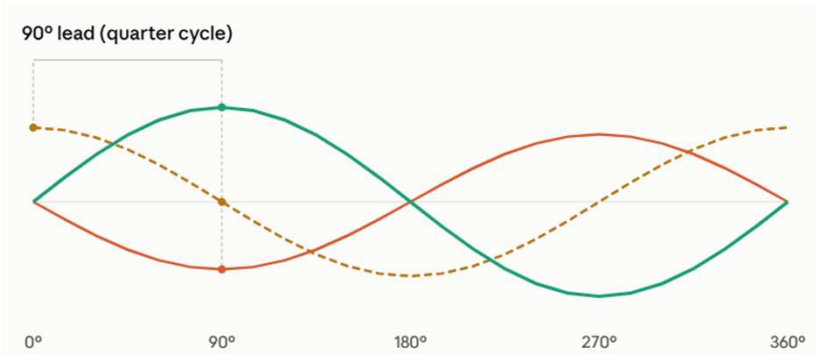
Therefore,

$$L_0 C_0 = \frac{1}{(3 \times 10^8)^2} = \frac{10^{-16}}{9}$$

Substituting,

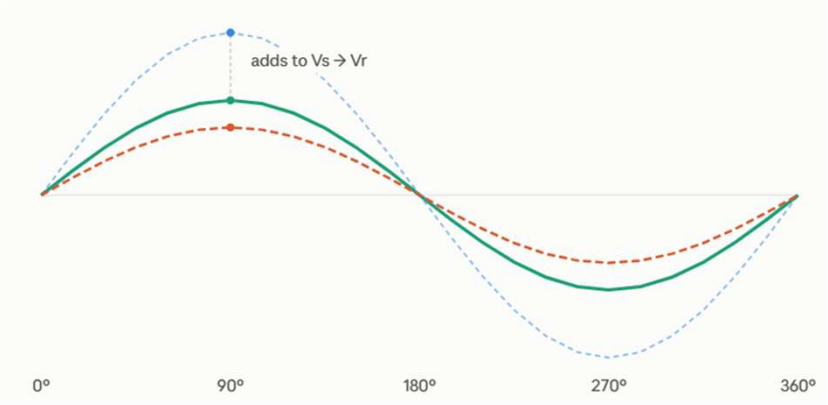
$$\boxed{\frac{V_R - V_S}{V_R} = \frac{\omega^2 l^2 \times 10^{-16}}{18}}$$

with l in metres.



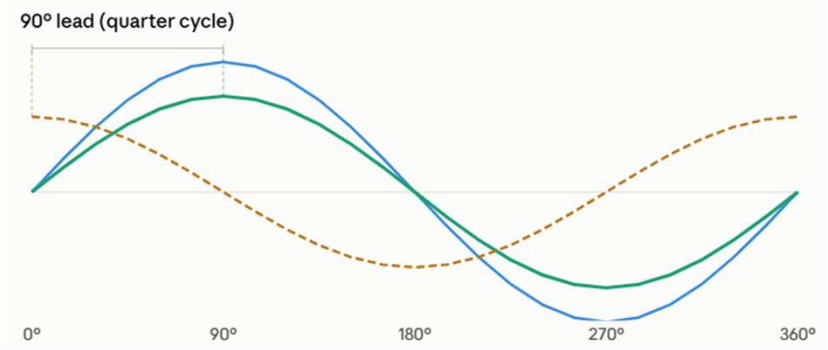
- Vs — sending-end voltage (base, 0°)
- Ic — charging current, leads Vs by 90°
- Ic × XL — the drop, antiphase with Vs

At 90°: Vs peaks while Ic×XL is at its most negative.



- Vs — sending-end voltage (base, 0°)
- −Ic×XL — now in phase with Vs
- Vr — the sum, same phase, larger peak

Vs and −Ic×XL peak together — they add, not cancel.



- Vs — sending-end voltage (base reference, 0°)
- Vr — receiving-end voltage, same phase, larger
- Ic — charging current, leads by 90°

Vr and Vs peak together — only the amplitude differs.

Effects and Mitigation of Ferranti Effect

Effects

- Receiving-end overvoltage under no-load or light-load conditions.
- Poor voltage regulation.
- Increased insulation stress on line equipment.

Methods to Reduce Ferranti Effect

- Install shunt reactors to absorb the charging current.
- Operate the line under suitable loading conditions.
- Use voltage control equipment in EHV transmission systems.

Example

A 300 km transmission line operating at 50 Hz may have a receiving-end voltage approximately 5% higher than the sending-end voltage under open-circuit conditions.

Example 6.22. Determine the percentage rise in voltage at the receiving end of a transmission line of length 200 km operating at 50 Hz.

Example 6.22. Determine the percentage rise in voltage at the receiving end of a transmission line of length 200 km operating at 50 Hz.

Solution : Rise in voltage $= \dfrac{\omega^2 l^2 V_R \times 10^{-10}}{18}$ volts

Percentage rise in voltage

$$= \frac{\omega^2 l^2 V_R \times 10^{-10}}{18 \times V_R} \times 100\%$$

$$= \frac{\omega^2 l^2 \times 10^{-8}}{18} = \frac{(2\pi \times 50)^2 \times (200)^2 \times 10^{-8}}{18} = 2.19\%$$ **Ans.**



Corona: The Invisible Glow

Corona is a localised, self-sustaining electrical discharge that occurs when the electric field at a conductor's surface exceeds the **critical disruptive voltage** — the threshold at which surrounding air ionises.

Visual Indicator

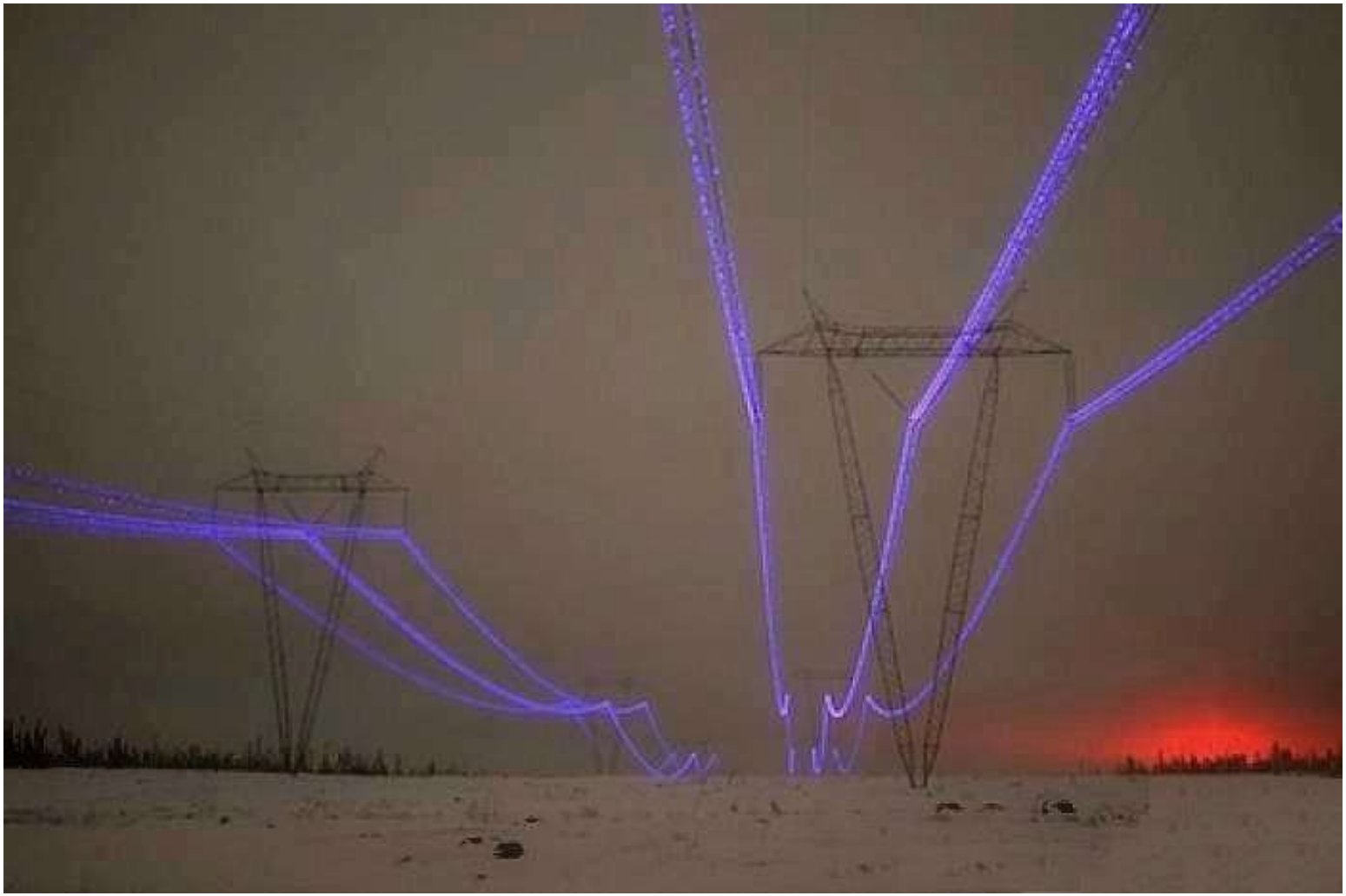
Faint violet or bluish glow surrounds the conductor, visible in darkness.

Audible Indicator

Distinctive hissing or crackling sound accompanies the discharge.

Chemical Indicator

Ozone (O_3) and oxides of nitrogen are produced, accelerating material degradation.



What is Corona?

Corona is the ionization of air surrounding a high-voltage conductor due to excessive electric field intensity.

Principle

- At low voltages, the air surrounding the conductor acts as an insulator.
- As the voltage increases, the electric field around the conductor also increases.
- Beyond a critical voltage, the surrounding air becomes ionized and starts conducting.
- This produces a faint violet glow, hissing noise, and ozone around the conductor.
- If the voltage is increased further, complete breakdown of air may occur, leading to flashover.

Corona is commonly observed in transmission lines operating at 100 kV and above.

Factors Affecting Corona

1. Atmospheric Conditions

• Corona is affected by the density and condition of the surrounding air.

• Humid, rainy or stormy weather reduces the breakdown strength of air, causing corona to occur at lower voltages.

2. Conductor Characteristics

• Larger conductor diameter reduces corona.

• Rough, stranded or dirty conductors increase corona due to higher electric field intensity at the surface.

3. Conductor Spacing

• Increasing the spacing between conductors reduces the electric field intensity and hence reduces corona.

4. Line Voltage

• It begins only when the electric field at the conductor surface exceeds the breakdown strength of air.

Advantages and Disadvantages of Corona

Advantages

- Increases the effective conductor diameter, reducing the electric field intensity around the conductor.
- Reduces the probability of flashover, thereby improving system reliability.
- Dissipates part of the energy associated with lightning and switching surges, reducing transient over-voltages.

Disadvantages

- Causes power loss, especially under humid and adverse weather conditions.
- Produces non-sinusoidal currents and voltages, causing interference with nearby communication lines.
- Produces ozone, which may chemically react with the conductor and cause corrosion.

Critical Voltages in Corona

As the line voltage is increased, corona develops in two stages:

Disruptive Critical Voltage (Vd)

- The voltage at which the electric field exceeds the dielectric strength of air.
- Air surrounding the conductor becomes ionized and corona just begins.

Visual Critical Voltage (Vv)

- The voltage at which the corona becomes visible as a faint violet glow accompanied by a hissing noise and ozone.

Corona starts at the disruptive critical voltage but becomes visible only after the visual critical voltage is reached.

Corona: Impact on Performance

Adverse Effects

- **Power loss:** Energy is dissipated in ionising the air — significant on 33 kV and above systems.
- **Radio interference (RI):** Electromagnetic emissions from corona disturb AM radio and communication signals.
- **Insulation degradation:** Prolonged corona leads to pitting, carbonisation, and eventual flash-over of insulation.

Design Countermeasures

Increase Conductor Diameter

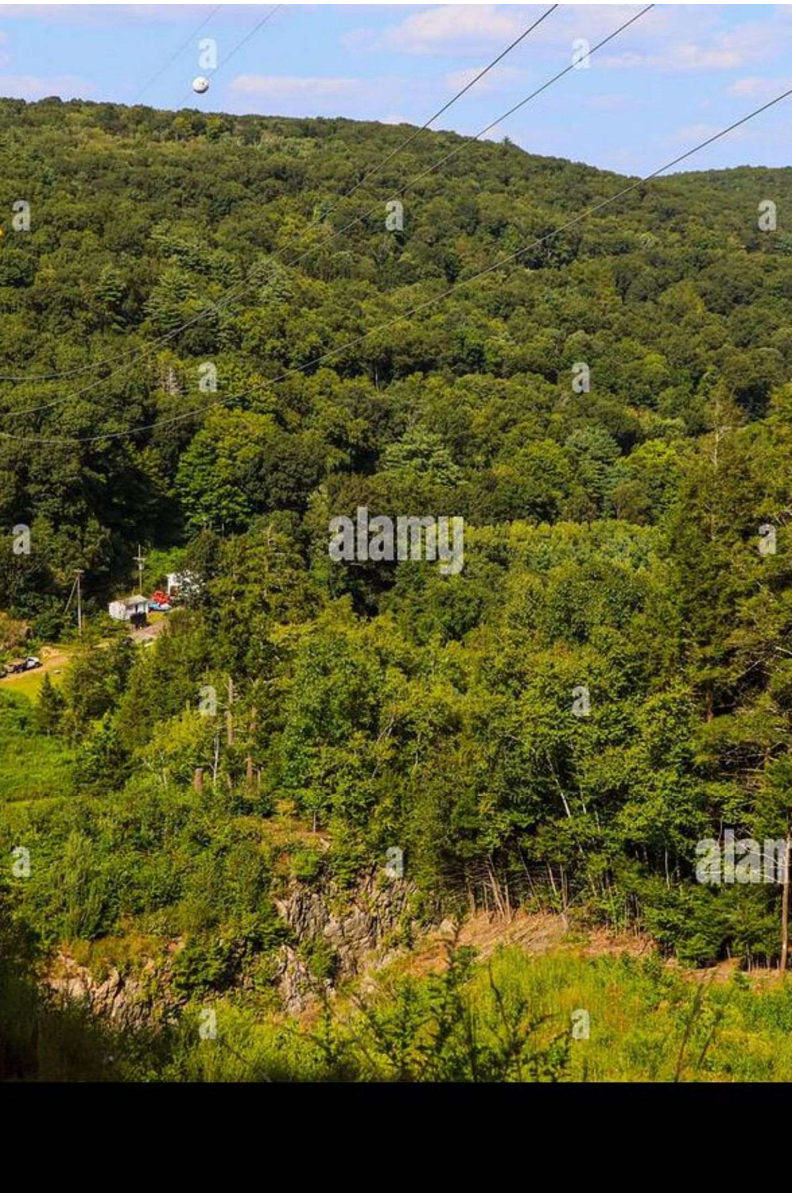
Reduces surface electric field gradient below the critical level.

Bundle Conductors

Multiple sub-conductors per phase effectively increase the equivalent diameter.

Smooth Conductor Surface

Eliminating sharp points and surface irregularities reduces localised field enhancement.



CHAPTER 3

Efficiency and Capacity

Zc

Surge Impedance

Characteristic impedance of a lossless line — **Zc = √(L/C)** — purely resistive in nature.

~400Ω

Typical Range

Overhead lines typically fall between **400 – 600 Ω**; cables are much lower (~40 Ω).

SIL

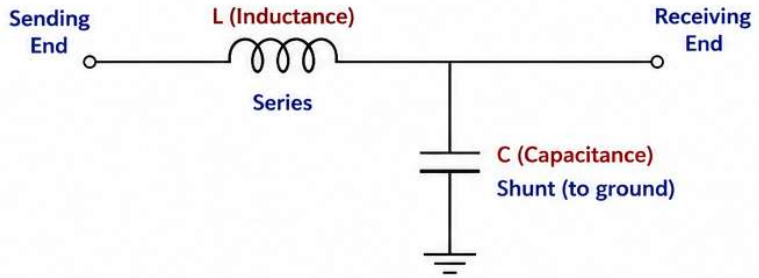
Natural Load

When a line is loaded at its SIL, reactive power is self-compensated — no net VAr exchange with the grid.

SURGE IMPEDANCE LOADING (SIL)

When the reactive power supplied by the line capacitance equals the reactive power absorbed by the line inductance, there is **no net reactive power** and the power factor becomes **unity**.
If the transmission line is loaded in such a manner, it is called **Surge Impedance Loading (SIL)**.

1. Transmission Line Model



- Inductance (L) is in **SERIES** with the line current.
- Capacitance (C) is in **PARALLEL** (shunt) with the line.

2. Reactive Power Balance at SIL

Reactive power supplied by line capacitance (Leading / Capacitive)	=	Reactive power absorbed by line inductance (Lagging / Inductive)
$Q_C = \frac{V^2}{X_C}$		$Q_L = \frac{V^2}{X_L}$
where $X_C = \frac{1}{\omega C}$		where $X_L = \omega L$

At SIL,

$$Q_C = Q_L \Rightarrow \text{Net reactive power} = 0 \Rightarrow \text{Power factor} = 1 \text{ (Unity)}$$

3. Surge Impedance Loading (SIL)

At SIL, the real power transferred is given by:

$$P_{SIL} = \frac{V^2}{Z_0} \text{ watts}$$

where,

$Z_0 = \sqrt{\frac{L}{C}}$ = Surge Impedance (ohms)
(Characteristic Impedance of the line)

4. Effect of Loading

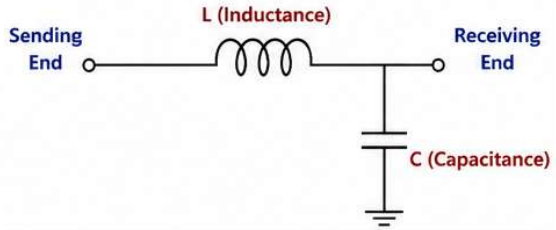
Loading Condition	Reactive Power	Effect on Voltage / PF
Load < SIL (Light Load)	$Q_C > Q_L$ (Capacitive)	Net leading reactive power Voltage rise (Ferranti Effect)
Load = SIL	$Q_C = Q_L$	No net reactive power PF = 1 (Unity), Flat voltage profile
Load > SIL (Heavy Load)	$Q_C < Q_L$ (Inductive)	Net lagging reactive power Voltage drop

Key Idea: At SIL, the natural leading vars from the line capacitance are exactly balanced by the lagging vars due to line inductance. Hence, the line draws no reactive power from the source and operates at unity power factor.

DERIVATION OF SURGE IMPEDANCE (Z_0)

At Surge Impedance Loading (SIL), the reactive power supplied by the line capacitance equals that absorbed by the line inductance. Hence, $Q_C = Q_L$.

1. Transmission Line Model (Nominal-π)



- Inductance (L) is in SERIES with the line current.
- Capacitance (C) is in PARALLEL (shunt) with the line.

2. Reactive Power Supplied by Capacitance (Q_C)

Shunt capacitance draws a leading (capacitive) reactive power given by

$$Q_C = \frac{V^2}{X_C} \quad \text{where} \quad X_C = \frac{1}{\omega C}$$

$$Q_C = V^2 \cdot \omega C$$

3. Reactive Power Absorbed by Inductance (Q_L)

Series inductance absorbs a lagging (inductive) reactive power given by

$$Q_L = \frac{V^2}{X_L} \quad \text{where} \quad X_L = \omega L$$

$$Q_L = \frac{V^2}{\omega L}$$

4. Condition at SIL (No Net Reactive Power)

At SIL, $Q_C = Q_L$

$$V^2 \cdot \omega C = \frac{V^2}{\omega L}$$

$$\Rightarrow \quad \omega^2 LC = 1$$

$$\Rightarrow \quad \boxed{LC = \frac{1}{\omega^2}}$$

5. Surge Impedance (Z_0)

Surge (characteristic) impedance of the line is

$$Z_0 = \sqrt{\frac{L}{C}}$$

Using $LC = \frac{1}{\omega^2}$ from Step 4,

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{1}{\omega^2 C^2}} = \frac{1}{\omega C} = X_C$$

6. Real Power at SIL

At SIL, current $I = \frac{V}{Z_0}$

$\therefore$ Real power transferred,

$$P_{SIL} = VI = \frac{V^2}{Z_0} \quad \text{watts}$$

Key Physical Insight

The line capacitance supplies leading reactive power, while the line inductance absorbs equal lagging reactive power. At SIL, they exactly balance, so the source supplies ZERO net reactive power and the power factor is UNITY.

Final Result

$$Z_0 = \sqrt{\frac{L}{C}} \quad \text{and} \quad P_{SIL} = \frac{V^2}{Z_0} \quad \text{watts}$$

(Surge / Characteristic Impedance) (Real Power at SIL)

When the line is loaded at P_{SIL}, it is called Surge Impedance Loading (SIL).

• Surge Impedance Loading (SIL) is the natural loading of a transmission line.
• It is the load at which the reactive power generated by the line capacitance is exactly equal to the reactive power absorbed by the line inductance.

Surge Impedance,

$$Z_0 = \sqrt{\frac{L}{C}}$$

L = Inductance per unit length
C = Capacitance per unit length
Natural (Surge Impedance) Loading

$$\text{SIL} = \frac{V^2}{Z_o} \text{ W per phase}$$

where V is the phase voltage.

$$\text{SIL} = \frac{V_{LL}^2}{Z_0}$$

Surge Impedance (Z_0): $\sqrt{\frac{L}{C}}$

where L is inductance, and C is capacitance per unit length

Below SIL: The line acts as a reactive power (MVAR) source, causing a voltage rise at the receiving end (Ferranti effect potential).

Above SIL: The line acts as a reactive power sink, consuming MVARs and causing a voltage drop.

At SIL: The voltage profile is flat; the sending-end voltage equals the receiving-end voltage, with zero net reactive power exchange.

Single Phase

$$(SIL)_{1\emptyset} = P_O = V_p I_p cos\emptyset$$

Since the load is purely resistive,

$$cos\emptyset = 1$$

$$P_O = V_P I_P = V_P \frac{V_P}{Z_O}$$

$$P_O = \frac{V_P^2}{Z_O} W/phase$$

3 Phase

$$Line\ voltage\ V_L = \sqrt{3} V_P$$

$$(SIL)_{3\emptyset} = 3P_O = \frac{3V_P^2}{Z_O} = \frac{V_L^2}{Z_O} W$$

If kVL is the receiving end voltage in kV, then

$$(SIL)_{3\emptyset} = \frac{(kV_L)^2}{Z_O} \text{MW}$$

A three-phase overhead transmission line has:

- Inductance per phase $L = 1.2\ \text{mH/km}$
- Capacitance per phase $C = 0.01\ \mu\text{F/km}$
- Line voltage $= 220\ \text{kV}$

Determine:

(a) The surge impedance of the line.

(b) The surge impedance loading (SIL) in MW.

(c) State whether the line will experience a voltage rise or voltage drop when operating at:

- (i) 80 MW
- (ii) 250 MW

$$L = 1.2 \times 10^{-3}\ \text{H/km}$$

$$C = 0.01 \times 10^{-6}\ \text{F/km}$$

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{1.2 \times 10^{-3}}{0.01 \times 10^{-6}}}$$

$$Z_0 = \sqrt{120000} = 346.4\ \Omega$$

$$\text{SIL} = \frac{V_{ph}^2}{Z_0}$$

$$V_{ph} = \frac{220}{\sqrt{3}} = 127\ \text{kV}$$

$$= \frac{(127 \times 10^3)^2}{346.4} = 46.5\ \text{MW}$$

Three-phase SIL $= 3 \times 46.5 = 139.5\ \text{MW}$

80 MW < 140 MW → **Voltage rise** (capacitive effect dominates).

250 MW > 140 MW → **Voltage drop** (inductive effect dominates).



CHAPTER 4

Mechanical and Structural Integrity

Conductors

ACSR or AAAC conductors carry current while withstanding mechanical tension, wind, and ice loading.

Towers

Lattice steel or concrete monopole structures maintain clearances and support conductor weight.

Insulator Strings

Disc insulators provide electrical isolation between energised conductors and earthed tower steelwork.

INSULATORS

Need for Insulators

- Overhead line conductors are bare and are not covered with insulating material.
- Insulators support the conductors on transmission towers and prevent current leakage to earth.
- They provide the required electrical insulation between the conductors and the supporting structures.

Functions of Insulators

- Prevent leakage current from the conductor to the tower.
- Maintain the required clearance between conductors and metal parts.
- Prevent short circuits between phase conductors.
- Provide mechanical support to the line conductors.

Insulators are one of the most important components of an overhead transmission system and their proper selection is essential for reliable operation.

Types of Insulators

The choice of insulator depends on the operating voltage, mechanical loading, and application.

The commonly used insulators are:

- Pin Insulator
- Suspension Insulator
- Strain Insulator
- Shackle Insulator

ED

Types of line Insulators



Glass Insulator



Disc Insulator



Post Insulator



Pin Insulator



Shackle Insulator



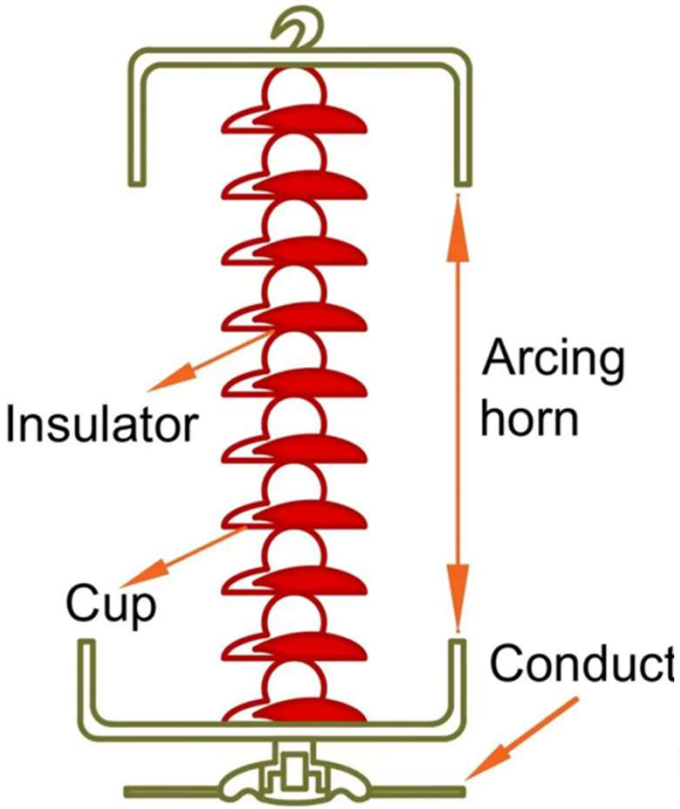
Strain Insulator



Stay Insulator



Suspension Insulator



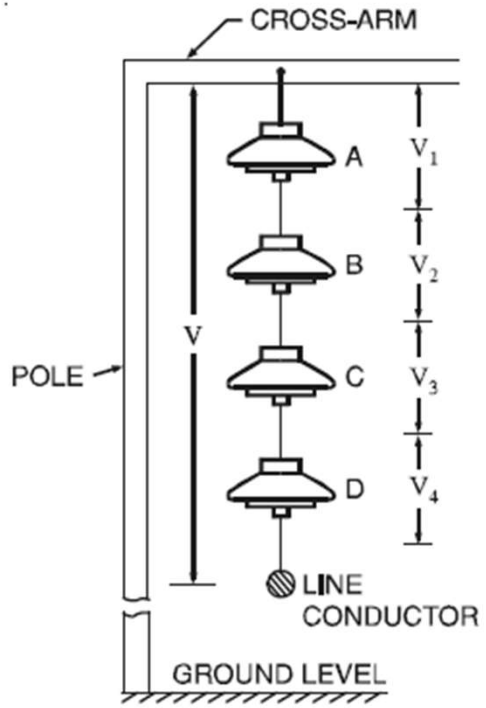


Fig. 3.15. *Arrangement of Insulator String*

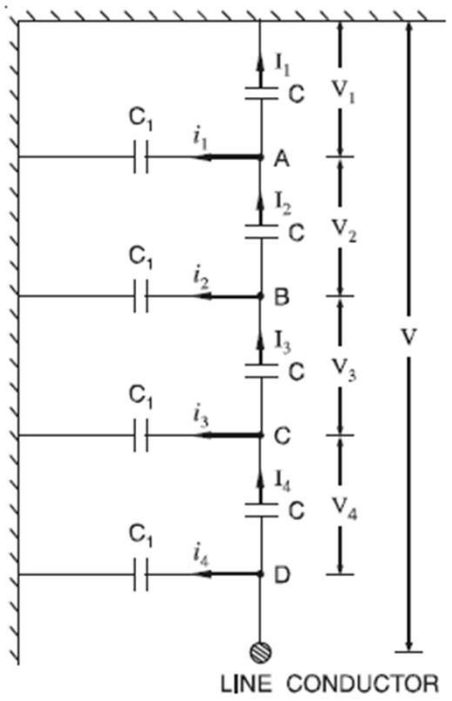


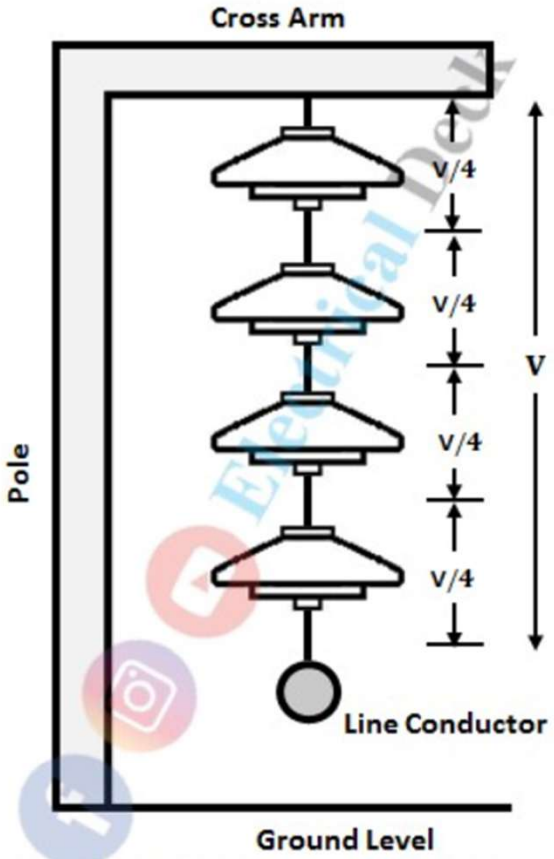
Fig. 3.16. *Equivalent Circuit For a 4-Unit String*

String Efficiency of a Suspension Insulator

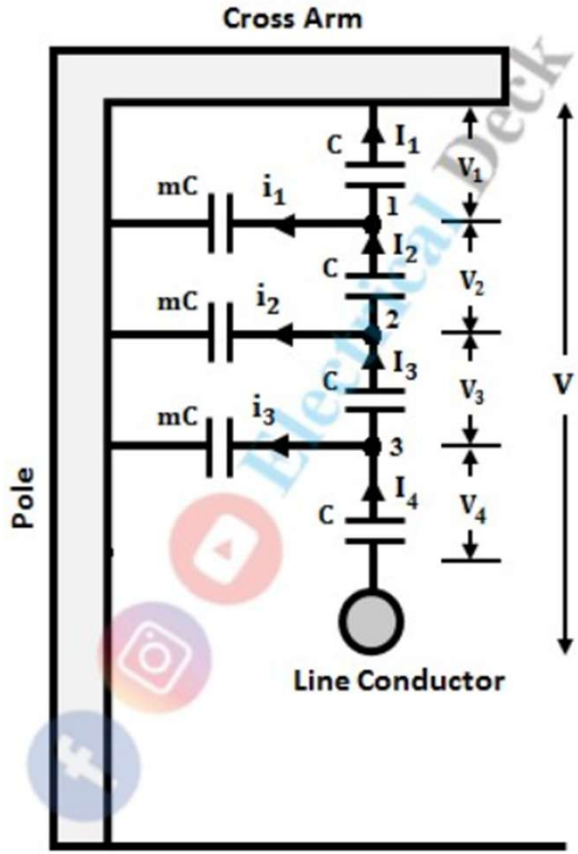
- Each insulator disc behaves as a capacitor of capacitance C.
- The metal links also have shunt capacitance kC to the earthed tower.
- Therefore, part of the current flows through the shunt capacitances.
- The current through each disc is not equal, causing unequal voltage distribution.
- The disc nearest the line conductor has the highest voltage stress.

String efficiency:

$$\eta = \frac{\text{Voltage across the string}}{\text{Number of discs} \times \text{maximum disc voltage}} \times 100\%$$



Suspension Insulator of 4 Disc



$$m = \frac{Earth\ capacitance\ or\ Capacitance\ to\ ground}{Self\ capacitance\ or\ Mutual\ capacitance}$$

Distribution of voltage at node 1 can be found by applying KCL,

$$I_2 = I_1 + i_1$$
$$V_2\omega C = V_1\omega C + V_1 m\omega C$$
$$\therefore V_2 = (1+m)V_1$$

Distribution of voltage at node 2,

$$I_3 = I_2 + i_2$$
$$V_3\omega C = V_2\omega C + (V_1 + V_2)m\omega C$$
$$V_3 = (1+m)V_1 + mV_1 + m(1+m)V_1$$
$$V_3 = V_1(m^2 + 3m + 1)$$
$$\therefore V_3 = V_1(1 + 3m + m^2)$$

Similarly, distribution of voltage at node 3,

$$I_4 = I_3 + i_3$$
$$V_4\omega C = V_3\omega C + (V_1 + V_2 + V_3)m\omega C$$
$$\therefore V_4 = V_1(1 + 6m + 5m^2 + m^3)$$

$$\eta = \frac{\text{voltage across the complete string}}{n \times \text{maximum voltage across any disc}} \times 100\%$$

$$\eta = \frac{V_1 + V_2 + V_3 + V_4}{4V_4} \times 100\%$$

For the standard capacitance model with shunt capacitance mC,

$$V_2 = (1 + m)V_1$$

$$V_3 = (1 + 3m + m^2)V_1$$

$$V_4 = (1 + 6m + 5m^2 + m^3)V_1$$

Hence, the total string voltage is

$$V = (4 + 10m + 6m^2 + m^3)V_1$$

Therefore,

$$\boxed{\eta = \frac{4 + 10m + 6m^2 + m^3}{4(1 + 6m + 5m^2 + m^3)} \times 100\%}$$

So the earlier equation was specifically for a three-disc string.

Memory Trick – Voltage Distribution Across Insulator String

For an ungraded string with up to 4 discs:

	1	m	m^2	m^3
V_1	1			
V_2	1	1		
V_3	1	3	1	
V_4	1	6	5	1

Easy Patterns

m coefficients → Triangular numbers:

$$\boxed{1,\ 3,\ 6}$$

$$1 \xrightarrow{+2} 3 \xrightarrow{+3} 6$$

Diagonal next to final 1 → Odd numbers:

$$\boxed{1,\ 3,\ 5}$$

$$1 \xrightarrow{+2} 3 \xrightarrow{+2} 5$$

Methods of Improving String Efficiency

The voltage distribution across a suspension insulator string can be made more uniform by:

1. Larger Disc Insulators

- Increase the self-capacitance of each insulator unit.
- Reduces the ratio of shunt capacitance to self-capacitance (m).

2. Longer Cross-Arms

- Increase the distance between the insulator string and the tower.
- Reduces the shunt capacitance to earth.

3. Capacitance Grading

- Use insulator units of progressively higher capacitance towards the line conductor.
- Produces a more uniform voltage distribution across the string.

4. Static Shielding (Grading Ring)

- A metal grading ring is connected to the line conductor.
- Redistributes the electric field and reduces the voltage stress on the conductor-end disc.

Capacitance Grading of an Insulator String

- In a suspension string, the conductor-end unit has the highest voltage stress.
- Capacitance grading uses insulator units of different capacitances to obtain nearly equal voltage across all units.
- The capacitance is made minimum at the tower end and progressively increased towards the line conductor.
- A larger capacitance gives a smaller capacitive reactance and helps reduce the voltage across the highly stressed lower units.
- Ideally, each unit is designed so that the voltage across every disc is the same.

Limitation

- Different-sized insulator units are required, making standardization, replacement and maintenance difficult.

Capacitance Grading – Four-Unit String

Let the voltage across each unit be v and the shunt capacitance be mC.

Capacitance of the top unit:

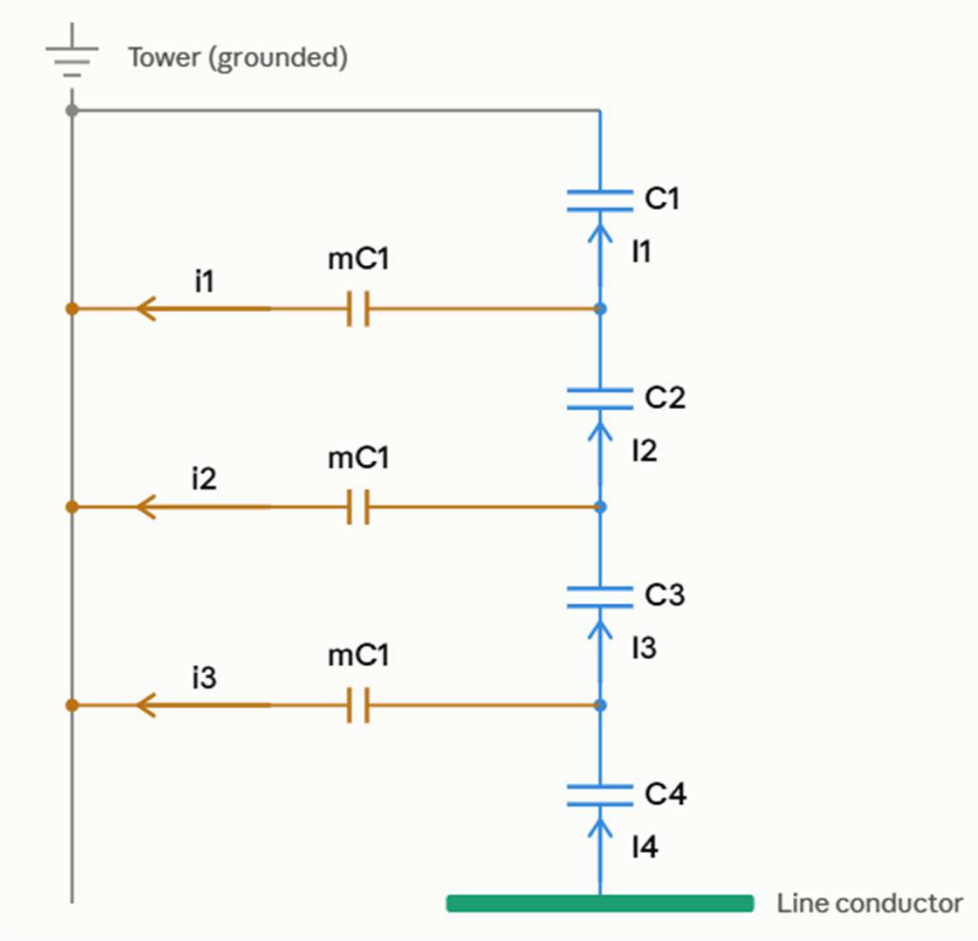
$C_1 = C$

Applying KCL successively at the junctions:

$C_2 = C(1 + m)$
$C_3 = C(1 + 3m)$
$C_4 = C(1 + 6m)$

Required capacitance ratio:

$C_1 : C_2 : C_3 : C_4$
$= 1 : (1 + m) : (1 + 3m) : (1 + 6m)$



Thus, capacitance must increase from the tower end towards the line conductor.

Memory Trick – Triangular Numbers

The coefficients of m follow:

$$\boxed{0,\ 1,\ 3,\ 6,\ 10,\ 15, \ldots}$$

because

$$0 \xrightarrow{+1} 1 \xrightarrow{+2} 3 \xrightarrow{+3} 6 \xrightarrow{+4} 10$$

Hence:

$$\boxed{C, \quad C(1+m), \quad C(1+3m), \quad C(1+6m), \ldots}$$

Capacitance increases towards the line conductor.

Example 3.4. A string of 4 insulators has self capacitance equal to 4 times the pin to earth capacitance. Calculate: (*i*) the voltage distribution across various units as a percentage of total voltage across the string, (*ii*) string efficiency.

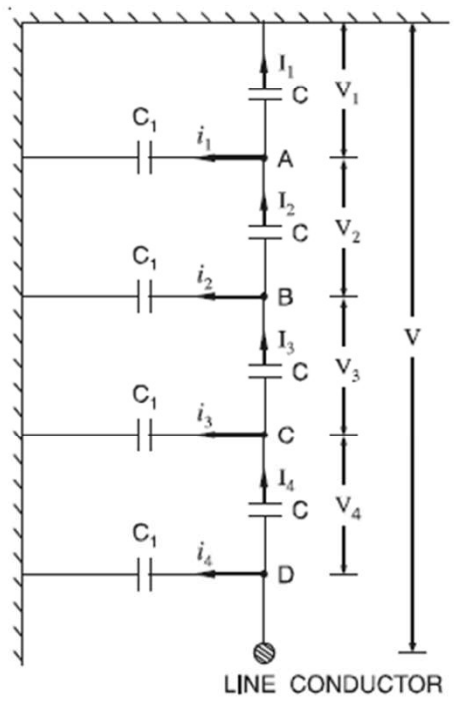


Fig. 3.16. *Equivalent Circuit For a 4-Unit String*

Solution : Ratio of shunt to mutual capacitance,

$$K = \frac{C_1}{C} = \frac{1}{4} = 0.25$$

Voltage across top unit,

$$V_1 = \frac{V}{4 + 10K + 6K^2 + K^3} \quad \text{... Refer to Eq. (3.6)}$$

$$= \frac{V}{4 + 10 \times 0.25 + 6 \times (0.25)^2 + (0.25)^3}$$

$$= \frac{V}{6.89} \text{ or } \frac{100}{6.89} \textit{ i.e.}, 14.51\% \text{ of total voltage across the string } \textbf{Ans.}$$

Voltage across the second unit from the top

$$V_2 = (1 + K)V_1 \quad \text{...Refer to Eq. (3.2)}$$

$$= 1.25 \times 14.51\% \text{ of } V$$

$$= 18.14\% \text{ total voltage across the string } \textbf{Ans.}$$

Voltage across the third unit from the top

$$V_3 = (1 + 3K + K^2)V_1 \quad \text{...Refer to Eq. (3.}$$

$$= (1 + 3 \times 0.25 + 0.25^2)V_1$$

$$= 1.8125 \times 14.51\% \text{ of } V$$

$$= 26.3\% \text{ of total voltage across the string } \textbf{Ans.}$$

Voltage across the fourth unit from the top

$$V_4 = (1 + 6K + 5K^2 + K^3)V_1$$

$$= (1 + 6 \times 0.25 + 5 \times 0.25^2 + 0.25^3)\, V_1$$

$$= 2.828 \times 14.51\% \text{ of } V$$

$$= 41.036\% \text{ of total voltage across the string. } \textbf{Ans.}$$

Percentage string efficiency,

$$\eta = \frac{V}{4V_4} \times 100 = \frac{V}{4 \times 0.41036\, V} \times 100 = 60.92\% \textbf{ Ans.}$$

Example 3.9. A string of 6 suspension insulators is to be graded to obtain uniform distribution of voltage across the string. If the pin to earth capacitance are all equal to C and the mutual capacitance of the top insulator is 10 C, find the mutual capacitance of each unit in terms of C.

Solution : Number of insulator units, $n = 6$

Capacitance to ground to top unit capacitance ratio,

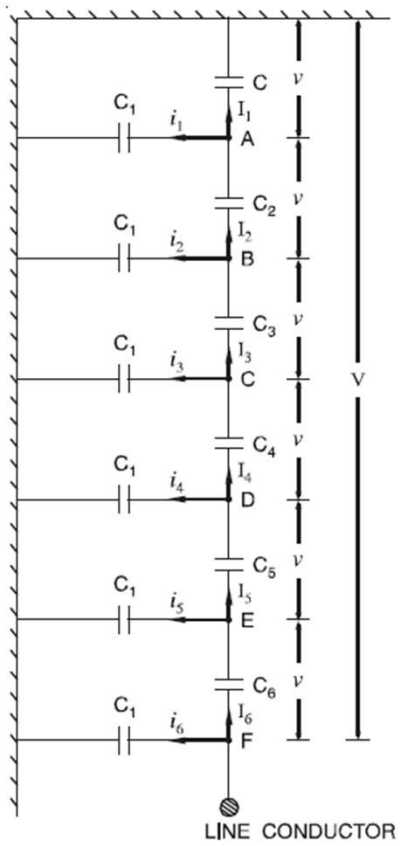
$K = 0.1$ or $C = 10\,C_1$

From equations (3.11), (3.12) and (3.13), we have

$C_2 = (1 + K)C = (1 + 0.1)C = 1.1 \times 10C_1 = 11C_1$ **Ans.**

$C_3 = (1 + 3K)C = 1.3C = 1.3 \times 10C_1 = 13C_1$ **Ans.**

$C_4 = (1 + 6K)\,C = 1.6C = 1.6 \times 10C_1 = 16C_1$ **Ans.**



Applying Kirchhoff's first law to node D, we have

$$I_5 = I_4 + i_4$$

$$\text{or } \omega C_5 v = \omega C_4 v + \omega C_1 \times 4v$$

$$\text{or } C_5 = C_4 + 4KC = (1 + 6K)C + 4KC$$

$$= (1 + 10K)C$$

$$= (1 + 10 \times 0.1)C = 2 \times 10C_1 = 20C_1$$ **Ans.**

Applying Kirchhoff's first law to node E, we have

$$I_6 = I_5 + i_5$$

$$\text{or } \omega C_6 v = \omega C_5 v + \omega C_1 \times 5v$$

$$\text{or } C_6 = C_5 + 5KC = (1 + 10K)C + 5KC$$

$$= (1 + 15K)C = (1 + 15 \times 0.1) \times 10C_1 = 25C_1$$ **Ans.**

String Efficiency

The Unequal Voltage Problem

In a string of disc insulators, voltage is **not distributed equally** across each disc. Mutual capacitance between each disc metal-work and the earthed tower causes more current to flow through discs nearer the conductor.

The disc **closest to the line conductor** therefore bears the highest voltage stress — and is most prone to flashover.

$$\eta = \frac{V_{string}}{n \times V_{disc_nearest_conductor}} \times 100\%$$

Where n is the number of discs. A perfect string would have $|eta = 100|\%$.

Typical Values

String efficiency ranges from **80 – 90%** in practice, depending on the ratio of mutual-to-self capacitance (K).

Effect of More Discs

Adding more discs to handle higher voltages *reduces* string efficiency further — making grading even more critical.

Grading the Insulators

Why Grade?

Grading aims to equalise the voltage across each disc, improving string efficiency towards 100% and extending the operational life of the insulator assembly.

Static (Capacitance) Grading

Metal fittings of increasing cross-sectional area are used on discs from top to bottom, intentionally varying the self-capacitance of each unit to equalise voltage drop.

Guard Rings

A toroidal metal ring fitted at the line-conductor end increases the capacitance of the bottom disc to the conductor, reducing the disproportionate voltage stress on the lowest disc and redistributing it upward.

Advanced Modelling Summary

Lumped Parameters

Short and medium lines — circuit constants concentrated at discrete points. Simple, fast, adequate for most planning studies.

1

2

Distributed Parameters

Long lines — R, L, C, G vary continuously. Hyperbolic functions (cosh, sinh) describe exact voltage and current profiles.

3

Simulation Tools

PSCAD, EMTP, and DIgSILENT PowerFactory implement full distributed-parameter models for transient and steady-state studies.

The engineering challenge is always to balance **model accuracy** against **computational complexity** — the right model is the simplest one that captures the phenomena of interest.

Current Trends in Transmission

Advanced Conductor Materials

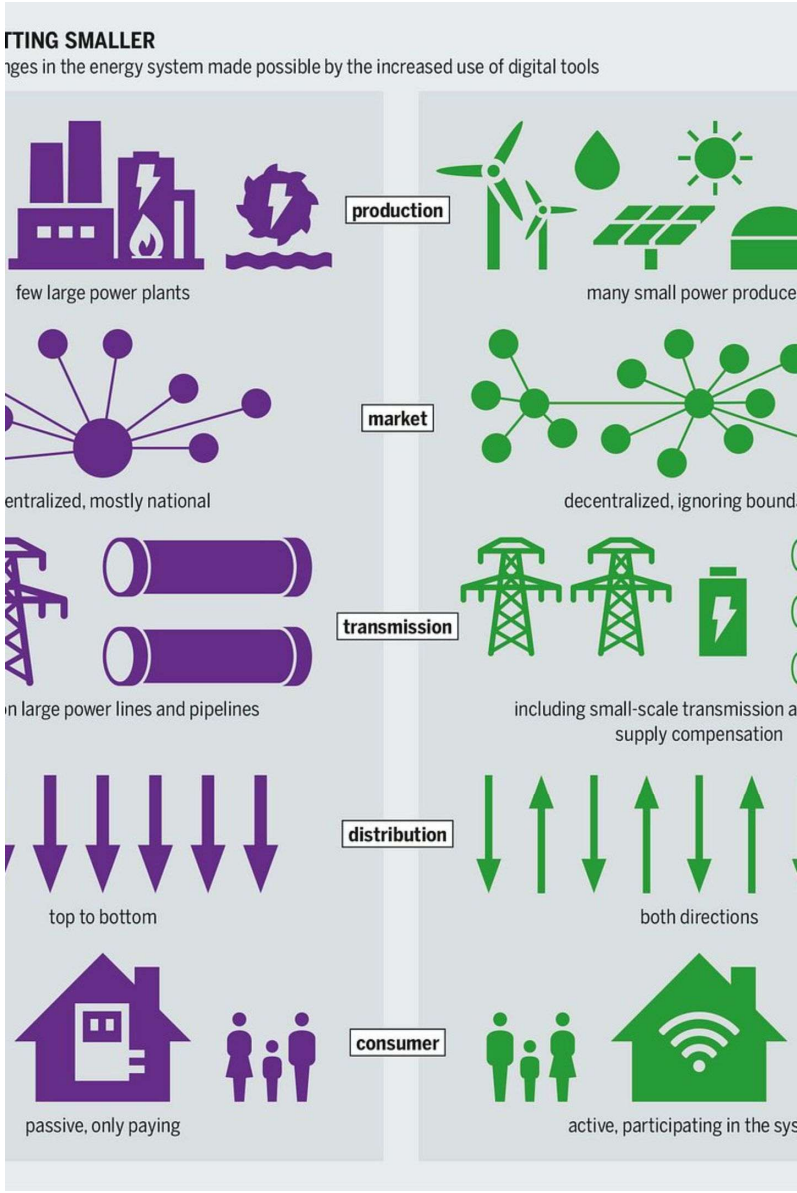
High-Temperature Low-Sag (HTLS) conductors such as ACCC and GTACSR allow significantly higher current ratings on existing towers without structural modification.

Real-Time Thermal Monitoring

Dynamic Line Rating (DLR) systems use weather and sag sensors to determine actual current capacity in real time — often 20–30% above static ratings.

Smart Grid and HVDC Integration

Voltage Source Converter (VSC) HVDC links enable long-distance bulk power transfer with full reactive power controllability — essential for integrating offshore wind and solar at scale.



Conclusion: Engineering for a Stable Future

Mastering the physics of the transmission line — from the mathematics of ABCD parameters to the subtleties of corona and insulator grading — is foundational to global energy security.

Accurate Modelling

Drives reliable voltage regulation, efficiency, and stability across all line lengths.

Phenomena Awareness

Understanding skin effect, Ferranti effect, and corona prevents costly design failures.

Design Optimisation

String efficiency, SIL loading, and conductor selection remain the primary defences against system collapse.